

# Water Treatment Site-Specific Operating Manual

SUMAS Environmental Services Inc. · Kamloops, BC

Living document — compiled for operators with no prior water treatment experience.  
Covers Sections 0–11. Glossary and the printable Logbook/Mixing Guide are separate companion documents.

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## Plant Orientation

### What This Plant Does

This is an industrial site water treatment system — not a municipal drinking water plant. Its job is to clean contaminated site water (stormwater, process water, and groundwater from the industrial site) before it's reused on site.

The source water has been in contact with industrial operations, so it typically carries:

- **Dissolved iron ( $\text{Fe}^{2+}$ )** — turns water orange/brown, clogs filters, must be oxidized and filtered out
- **Petroleum hydrocarbons** (oils, fuels) — toxic to aquatic life, requires oil/water separation
- **High suspended solids / turbidity** — must be settled and filtered out
- **pH imbalance** — affects every downstream chemical step
- **Fine silts and colloidal particles** — too small to settle on their own, need coagulant and flocculant to clump together

**Treated water goes to the on-site storage tanks (used for dust control). This site has no permit to discharge to the drain — your job ends once water reaches the storage tank.**

The main operational target at every stage is getting the water **visually clear**. The carbon vessel at the end of the line will catch what earlier steps miss, but starting clear protects that last line of defense — and clear water is what the dust-control reuse actually needs.

### The Treatment Train, Start to Finish

Water moves through these stages in order:

1. **Open pond** — gravity pre-settling; heavy solids sink, oil floats. No equipment, just time.
2. **Oil/Water Separation Cell** — oil is skimmed off as waste. Flocculant is dosed here too (the first dosing point).
3. **Frac tanks** — outdoor storage tanks where water settles further; clearer water is drawn off from below.
4. **C Can (chemical dosing station)** — check pH; add lime if below 8; add PACI coagulant; add flocculant (always at least the coagulant dose).
5. **Outdoor weir tank / blue bin** — settles until it looks as clear as tap water.
6. **Ozone, then pump** — ozone generator turned on first, then the submersible pump pulls water (mid-depth) into the indoor intake pipe, where ozone is injected in transit.
7. **Indoor settling bins / wet cell** — settles again, same clarity check.
8. **Bag filter** — first filtration step the water actually hits (the sand filter on site is not used).
9. **Carbon vessel (SRS-CV-15)** — activated carbon, the last line of defense for dissolved fuel smell/colour.
10. **Storage tanks (Tank 1 / Tank 2)** — treated water lands here and is also used for on-site dust control. This is where your responsibility for the water ends.

Sludge management runs in parallel, not as a step in the main line — it's your actual last task each cycle: an AODD diaphragm pump sends settled sludge from the settling bins/weir tank to the Kontek filter press after treated water has already moved on to the storage tanks.

**Terminology:** "The lagoon" is just another name for the Oil/Water Separation Cell (step 2). There is no separate final-disposal lagoon — the job ends at the storage tank.

## Outdoor: Pond to Frac Tanks

1. Raw dirty water collects in the open pit/pond. Oil floats, solids sink — gravity does the work, no equipment.
  2. In the Oil/Water Separation Cell — a concrete two-sided channel — floating oil (a dark maroon/red liquid) is skimmed off and disposed of as waste; it never reaches the clean side. Flocculant (CL-200 polymer) is added here too.
  3. Frac tanks (~100,000 L portable steel tanks) give more settling time. Clearer water is drawn from the middle/below. Tanks are labelled clean / dirty / in treatment — know what's in each one before moving anything.
- Oil is removed first because it's toxic to discharge and fouls everything downstream.

## How Water Actually Moves

Nothing between the outdoor tanks is permanently plumbed. A submersible pump (motor and pump sealed in one waterproof housing) is physically carried to whichever tank needs to be emptied, dropped in fully underwater, with a 2" hose run to the receiving tank. The same pump and hose setup is reused between the frac tank and the weir tank. Once the pump is sitting in a tank, all you'll see from outside is the hose fitting where it connects — that's normal.

**Check before assuming:** water doesn't move on its own here — confirm a pump is actually sitting in a tank right now, and trace where its hose goes.

## The C Can: Dosing the Chemicals

Water is pumped from the frac tank into the C Can, parked beside the weir tank. Inside, three drums hold the stock chemicals: far left = flocculant (white drum), middle = PACl coagulant (white drum), right = lime (blue drum).

1. Check the pH first — below 8 is a problem for the next steps.
2. If pH is under 8, add lime (quicklime/CaO). Only added when the reading actually comes back low.
3. Add PACl coagulant to neutralize the charge on fine particles so they stick together.
4. Add flocculant (CL-200) right after, to net the coagulated particles into heavier clumps that settle.

**Rule:** the flocculant dose must always be equal to or more than the coagulant dose — never less.

## Outdoor Weir Tank (Blue Bin): Settling

Dosed water flows from the C Can into the outdoor blue weir tank, where it sits and settles. There is no fixed settling time — it's visual: water should look as clear as tap water before moving on. Without the chemical dosing from the C Can, this tank does almost nothing on its own; it only drops the biggest particles.

Sludge collects on the tank floor over time and needs the diaphragm (AODD) pump to remove it — a regular submersible pump would clog on it.

## Ozone, Then Pump Indoors

1. Turn the ozone generator on first.
2. Then start the submersible pump, placed mid-depth in the weir tank — avoiding sludge on the bottom and the oil layer on top.
3. As water is pulled through the intake pipe into the building, ozone is injected into it along the way.

Ozone is a strong oxidizer — it breaks down dissolved hydrocarbons and converts dissolved iron into a solid form while the water is in transit.

**Order matters:** ozone on first, then the pump. Starting the pump first pulls untreated water in before ozone is doing anything.

## Indoor Settling, Then Filtration

Water arrives indoors through an overhead PVC pipe and settles again in the indoor weir tank/settling bins — same clarity check as outdoors. Once settled, a submersible pump pushes it onward through the filters to the storage tank.

Filtration runs through two stages only (the sand filter on site is not used):

1. **Bag filter** — a 100-micron pre-filter bag, then a 10-micron polishing bag. Since the sand filter isn't used, this is catching everything ahead of carbon, so bags need checking and changing more often.
2. **Carbon vessel (SRS-CV-15)** — activated carbon that grabs dissolved fuel smell/colour that filters can't catch.

**Turbidity (NTU) is checked at both the bag filter outlet and the carbon vessel outlet — these are the site's real compliance check points.**

Treated water that passes both filters goes to Tank 1 / Tank 2 — two large white storage tanks, each with its own ladder. This water also supplies on-site dust control (sprayed on roads/yard).

The carbon vessel (SRS-CV-15) is a blue pressure vessel rated to 75 PSIG maximum. It's backwashed weekly as routine maintenance and checked every 1.5 months for exhaustion — if smell or colour starts coming through again, the carbon needs replacing; backwashing can't fix exhausted carbon.

## Your Last Task Each Cycle: Sludge Removal

The job ends once treated water is pumped into the storage tank. What happens to the water after that point isn't tracked here.

The actual last task in the cycle is cleanup: once treated water has been pumped out of the indoor settling bins into the storage tank, use the AODD diaphragm pump to pull the settled sludge out of the settling bins and the outdoor weir tank, and send it through the Kontek filter press.

## Why the Orange Staining Is Normal

Heavy orange/rust staining on concrete surfaces throughout the facility — settling tanks, wet cell walls, floors, pipe connections — is normal for this site and not a sign of neglect.

The source water contains high levels of dissolved iron ( $\text{Fe}^{2+}$ ), which is invisible in solution but turns orange-brown the moment it's oxidized. When that oxidized iron precipitates out as iron hydroxide, it coats everything it touches with the characteristic rusty stain. The staining is simply a record of years of iron removal doing its job.

## Site Reference

|                          |                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------|
| Site type                | Industrial/commercial site water treatment                                                    |
| Associated operation     | Shell Canada petroleum operations                                                             |
| Location                 | Kamloops, BC Interior                                                                         |
| Water treatment supplier | SUMAS Environmental Services Inc. — 1-250-374-4151                                            |
| Regulatory framework     | BC Environmental Management Act                                                               |
| Discharge permit         | None — site has no permit to discharge to the drain. Treated water goes to the storage tanks. |

**Testing limitation:** this site does not have equipment to test for dissolved contents — testing relies on visual clarity/turbidity rather than on-site lab analysis of dissolved analytes.

## How the Plant Actually Runs Day to Day

The operating cycle, in plain terms:

1. Consolidate dirty water — move raw/untreated water into the right tanks for processing.
2. Filter the wet cell water — push water from the wet cell through the bag and carbon filter to the clean frac tank.
3. Empty the wet cell — get it as dry as possible so accumulated sludge/solids can be pumped out.
4. Treat and store — dose chemicals, run the full treatment train, fill the clean storage tanks, collect a compliance sample.
5. Move oil and dirty residuals — transfer separated oil and dirty wet cell water to the stabilization pit for further management.

This system runs in batch mode, not continuous flow. Knowing which tanks hold what type of water, and in what order to process them, is the core operational skill on this site.

**"Make sure the flow coming in isn't too much or there won't be enough time for settling." This is a balancing process, not an automated one — watching, adjusting, and using judgment at every step is the job.**

## SECTION 1

# Daily Operations

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### The Golden Rule of Shift Work

**Leave the plant in better shape than you found it.**

Before you leave, your relief operator should be able to open the logbook and understand exactly what happened on your shift — and what to watch for on theirs.

### Shift Start: Read the Handoff First

Before touching any equipment, pick up the logbook and any task notes from the previous shift and read every entry. Look for:

- Any equipment that is offline, locked out, or bypassed
- Frac tank levels and status (clean / dirty / in treatment)
- Any chemical shortages flagged
- Ongoing treatment batches in progress, and what stage they're at
- Unusual water quality (high turbidity, pH out of range, petroleum odour in effluent)
- Tasks carried over to your shift

**Why this matters:** Water is staged across multiple tanks at different treatment stages, and there's no centralized monitoring system tying it together. If you skip the handoff, you won't know where each batch is — and you could mix clean water with dirty water.

### Visual Check: Control Panel and Equipment

Walk to each piece of powered equipment and confirm its status. There is no centralized monitoring here — you are the monitoring system, so every item below has to be checked individually, every shift.

1. PACI dosing pump (blue) — running or standby per the dosing plan
2. CL-200 flocculant pump (yellow LMI) — running or standby per the dosing plan
3. CD30TIX ozone generator — ON indicator lit, pressure gauge normal
4. AERQUS ozone diffuser — bubbling in the contact water
5. Bag filter housing — flow OK, bag not overloaded
6. Kontek filter press — status matches the previous shift's note, if in cycle
7. H<sub>2</sub>O<sub>2</sub> drums — note the level in the log
8. Cal-Hypo tablets in the chlorinator — confirm not empty

## Physical Rounds: Walk the Whole Site

Walk the full site in the same order every time, and use all your senses — sight, smell, and sound all matter.

**Outdoor:** settling pond levels, any oil sheen or debris blocking the outlet; frac tank levels and whether their clean/dirty/in-treatment labels are current; any visible petroleum around the oil/water separator housing; any leaking hoses or connections.

**Indoor chemical/treatment room:** any chemical spills on the floor; any strong chemical smell beyond faint ozone or chlorine; cracked, kinked, or spraying pump tubing on the PACI and CL-200 pumps; drips or leaks at the bag filter housing lid seal; petroleum odour from the carbon vessel effluent; sludge buildup or clarity in the wet cell.

**Smell test:** Petroleum/fuel smell in treated water means the carbon vessel is likely exhausted. Rotten-egg ( $\text{H}_2\text{S}$ ) smell means possible biological activity in an anaerobic sludge pocket. Strong bleach smell means possible Cal-Hypo over-dose or spill. A sharp metallic/bleach smell is normal near the ozone system — but if it's noticeable elsewhere, ventilate.

Record everything you observe on rounds, whether it looks normal or not.

## Daily Samples

Run these tests every shift, immediately after collecting each sample, and record results in the log with the time taken:

1. Settling pond outlet / wet cell inlet — pH, turbidity (visual check is fine in the field)
2. After chemical dosing (wet cell outflow) — pH, turbidity
3. Bag filter outlet — pH, turbidity (NTU)
4. Carbon vessel effluent — pH, turbidity, petroleum odour check
5. Final treated water, pre-discharge — pH, free chlorine, turbidity. This is your compliance sample.

The oil/water separator also gets a weekly dip check (fuel thickness, sludge thickness, liquid level), logged separately.

## Continuous Monitoring Through the Shift

Every 1–2 hours, work back through the same checks:

1. Chemical drum/tank levels — are the dosing pumps consuming at the expected rate?
2. Frac tank levels — update status labels if any transfers happened
3. Bag filter housing — is flow rate holding, or is the bag loading up?
4. Ozone system — CD30TIX indicator still on, pressure gauge normal?
5. Free chlorine in treated water — confirm the residual is maintained
6. Log everything, even when there's no change

## If a Reading Is Out of Range

1. Confirm it's real — re-test manually, don't trust a single reading
2. Protect the discharge point first — if treated water quality is in question, stop the discharge valve
3. Find the cause (see Troubleshooting)
4. Fix it, or escalate to your supervisor
5. Record the time of the problem, the reading that flagged it, what you did, and the result

**Why this matters:** Regulators can inspect at any time and ask to see your records. A discharge that exceeds permit limits and has no log entry looks far worse than one that was caught, logged, and corrected.

## Key Parameters and Targets

Operational targets below are typical for industrial discharge treatment — confirm your site's actual permit limits with your supervisor, since those govern.

| Parameter                          | Typical Discharge Limit | Operational Target | Why It Matters                                              |
|------------------------------------|-------------------------|--------------------|-------------------------------------------------------------|
| Turbidity (final discharge)        | ≤ 25 NTU                | ≤ 5 NTU            | Turbidity carries other contaminants and harms fish habitat |
| pH (final discharge)               | 6.5–9.5                 | 7.0–8.5            | Too acidic or alkaline is toxic to aquatic life             |
| Total suspended solids (TSS)       | ≤ 25–75 mg/L            | Minimize           | Settled solids can smother fish spawning areas              |
| Free chlorine (final discharge)    | ≤ 0.1 mg/L              | ≤ 0.05 mg/L        | Toxic to fish even at low levels                            |
| Total petroleum hydrocarbons (TPH) | ≤ 1–5 mg/L              | Not detectable     | Petroleum is toxic to aquatic life                          |
| Iron (total)                       | ≤ 0.3–1 mg/L            | Minimize           | Iron precipitation in receiving waters harms fish           |

## Shift End: Last 30 Minutes

Before your relief arrives:

1. Complete all log entries and test results — no blanks
2. Note any equipment that is offline, in manual, or acting abnormally
3. Check chemical inventory and flag anything running low
4. List any tasks you couldn't complete, for the next shift
5. Prepare a short verbal briefing for your relief — five minutes face-to-face is worth more than a page of notes

Brief the incoming operator on what happened during your shift, the current status of all major processes, anything to watch closely, and any ongoing issues or anomalies.

**Sign the logbook:** the moment you sign, you own that shift's record.

## SECTION 2

# Startup Procedures

**⚠ Generic template — not yet site-confirmed:** the steps below describe a conventional municipal drinking-water plant startup (raw water intake, flocculation basin, clear well, distribution system). This site's actual process (pond → separator → frac tanks → C Can → ozone → indoor settling → bag filter → carbon vessel) has not yet been mapped to a startup sequence. Treat this section as general background only until confirmed with your supervisor.

### Types of Startup

There are three kinds of startup, depending on how long the plant has been offline:

1. **Cold start** — the plant has been fully offline (power outage, extended maintenance). Every system starts from scratch. This is the longest and most methodical type.
2. **Warm start** — the plant was only offline a few hours and systems are still primed. Shorter checklist.
3. **Unit process restart** — a single unit (one filter, one clarifier) is coming back online after maintenance. Only that unit's checklist applies.

**Not sure which type applies?** Treat it as a cold start. It's always the safer choice.

### Pre-Startup Checklist

Complete this before turning anything on, regardless of startup type.

**Safety:** all lockout/tagout (LOTO) locks removed by the person who placed them; no personnel inside basins, tanks, or vessels; all manhole covers, hatches, and access points closed and secured; safety shower and eyewash stations operational; logbook reviewed for open work orders or known issues.

**Mechanical:** pump packing glands / mechanical seals inspected for leaks; valve positions confirmed against the startup configuration on your P&ID; motor coupling guards in place; chemical injection points clear of blockages.

**Chemical:** chemical tanks stocked for at least one full day of operation; dosing pump tubing checked for cracks or kinks; dry dosing pumps primed (see Priming Chemical Dosing Pumps below).

**Electrical / instrumentation:** control panels energized; online instruments (turbidimeters, chlorine analyzers, flow meters) reading and fault-free; SCADA communications live, if applicable.

## Cold Start Sequence

7 phases, raw water to finished water — work through them in order, never skip ahead

**Phase 1 — Raw Water Intake.** Slowly open the raw water intake valve (no more than 25% initially) and start the raw water (low-lift) pumps at minimum speed/flow. Confirm flow on the raw water meter and wait for it to stabilize before proceeding. Watch for air release at high points in the suction line and open air release valves as needed.

Starting at low flow prevents water hammer — a pressure surge that can crack pipes and damage valves. Letting trapped air escape prevents air locks and inaccurate flow readings.

**Phase 2 — Chemical Dosing (Coagulant).** Start the coagulant dosing pump (typically alum or polyaluminium chloride/PACl) and set the dose to your baseline starting point from the dosing table. Confirm the pump is actually injecting by watching the injection point for the chemical plume, or verifying the tank level is dropping.

Coagulant must be dosing **before** water enters the flocculation basin. If water passes through without it, suspended solids carry straight to the filters — and a resulting turbidity spike can take an hour or more to clear.

**Phase 3 — Flocculation.** Start flocculation mixers at low speed and observe the floc forming in the basin. Healthy floc looks like small white or grey snowflakes that grow and settle; poor floc is pinpoint-sized or stays in suspension. Flocculation uses gentle mixing to help coagulated particles collide and stick together into larger, heavier floc that can settle out.

**Phase 4 — Sedimentation / Clarification.** Confirm the sedimentation basin outlet valve is open. As flow enters the basin, watch for even distribution across the inlet — uneven flow causes short-circuiting, where water bypasses settling and exits too fast. Do not expect good settled-water quality for the first 1–2 hours of startup; the basin needs time to reach steady state.

**Phase 5 — Filtration.** Before opening a filter to flow, confirm it has been backwashed within the last backwash cycle — if not, backwash it first. Slowly open the inlet valve (no more than 25% initially) and monitor filter effluent turbidity continuously. Turbidity may run elevated for 10–30 minutes as the filter "ripens."

Do not send filter effluent to the distribution system until turbidity is below your target — typically  $\leq 0.3$  NTU, often  $\leq 0.1$  NTU per your permit.

**Filter ripening is normal:** the filter media needs time to re-establish its biological and physical filtering layer. Send the first-flush water to waste or back to the head of the plant — never to distribution.

**Phase 6 — Disinfection.** Start the disinfection system (chlorine dosing pump, UV system, or both, per your plant design) and set the dose to your target CT value (contact time  $\times$  concentration).

**Before distribution:** confirm finished water turbidity is at or below your permit limit AND free chlorine residual is at or above the regulatory minimum at the entry point to distribution. Sending under-disinfected water to the distribution system is a public health emergency. If you are unsure, recirculate or send to waste, and call your supervisor.

**Phase 7 — Ramp Up to Normal Flow.** Gradually increase raw water flow to the operational target over 30–60 minutes, adjusting chemical doses proportionally as flow increases. Take a full set of startup samples — raw, settled, filter effluent, finished water — and record them in the logbook. Confirm all online instruments agree with your grab samples within  $\pm 10\%$ .

## Unit Process Restart

When bringing a single unit back online (for example, a filter or clarifier that was taken offline for maintenance):

1. Confirm the maintenance work order is signed off and LOTO removed.
2. Run the relevant portion of the Pre-Startup Checklist.
3. Bring the unit online slowly while keeping the rest of the plant at stable operation.
4. Monitor that unit's effluent quality for 30 minutes before returning it to normal service.
5. Record in the logbook: time offline, reason, and time returned to service.

## Priming Chemical Dosing Pumps

If a dosing pump has run dry (empty tube or tank), prime it before use:

1. Make sure the chemical tank has liquid in it.
2. Disconnect the discharge tubing from the injection point, holding a rag over it to catch any chemical.
3. Run the pump until chemical flows steadily from the discharge with no bubbles.
4. Reconnect the tubing.
5. Verify the pump is injecting correctly at the injection point.

**PPE required:** wear gloves, goggles, and a face shield when working with dosing chemicals.

## Startup Completion Checklist

Before declaring startup complete, confirm all of the following:

1. All processes are running at target flow.
2. Chemical doses are set and confirmed injecting.
3. Finished water turbidity is within target.
4. Finished water chlorine residual is within target.
5. Finished water pH is within target.
6. All online instruments are reading and calibrated.
7. Startup is logged with start time and initial readings.
8. Supervisor is notified if this was an unplanned startup.

## SECTION 3

# Shutdown Procedures

**⚠ Generic template — not yet site-confirmed:** the steps below describe a conventional municipal drinking-water plant shutdown (raw water pumps, flocculation/sedimentation basins, clear well, distribution entry valve). This site's actual batch process has not yet been mapped to a shutdown sequence. Treat this section as general background only until confirmed with your supervisor.

### Types of Shutdown

There are three kinds of shutdown, and the right one depends on how much warning you have:

1. **Planned shutdown** — scheduled maintenance, holiday, or seasonal. Low urgency, time to prepare.
2. **Controlled emergency shutdown** — equipment failure, power issue, or contamination event. Medium urgency — act within minutes.
3. **Emergency stop (E-stop)** — immediate threat to human safety. Act instantly.

### Planned Shutdown — Before You Start

Hours before a planned shutdown:

1. Notify your supervisor and any downstream water users if supply will be interrupted.
2. Fill the clear well / storage reservoir to maximum capacity — this buffers supply during the shutdown.
3. Confirm chemical inventory is sufficient for restart.
4. Document the current state of all processes (flow rates, doses, pressures) as your baseline to return to.

### Planned Shutdown — Sequence

**Always shut down in reverse order — from distribution back to intake — so you never push untreated water forward.**

1. Reduce raw water flow gradually over 15–30 minutes. Don't shut off suddenly — this causes water hammer.
2. Reduce chemical doses proportionally as flow decreases.
3. Stop the raw water pump(s) and close the intake valve.
4. Stop coagulant and any pre-treatment chemical dosing pumps.
5. Let flocculation and sedimentation basins drain to waste, or leave them filled, per your SOP — keeping basins full prevents structural damage from differential pressure on the walls.
6. Close filter inlet valves and note filter head levels.
7. Reduce chlorine dosing to zero, or maintain a minimal dose if water will sit in the contact chamber for an extended period.
8. Close the entry point valve to the distribution system — only once the clear well level is sufficient for the shutdown period.
9. Record shutdown time, reason, and plant state in the logbook.
10. Apply lockout/tagout (LOTO) to any equipment that maintenance will work on.

### Planned Shutdown — After You're Done

1. Walk the plant and confirm all equipment that should be stopped is stopped.
2. Secure chemical rooms — locked and ventilated.
3. Confirm the clear well level is logged.
4. If you'll be away, post a "PLANT SHUTDOWN — DO NOT OPERATE" notice on the main panel.

### Controlled Emergency Shutdown

Triggers include major equipment failure (pump cavitation, motor failure), chemical feed system failure, a pipe break, or a power fluctuation. Move faster than a planned shutdown, but still work through the steps in order:

1. Immediately assess whether the problem is affecting finished water quality. If yes, close the distribution entry valve right away, then troubleshoot. If no, you may have time for an orderly shutdown.
2. Stop the raw water pump(s).
3. Stop chemical dosing.
4. Record the time, the triggering event, and all current readings in the logbook.
5. Notify your supervisor immediately.
6. Follow emergency response procedures if the event involves a chemical spill, contamination, or safety hazard.

### Emergency Stop (E-Stop)

Use the E-stop only when there is an immediate threat to human safety — fire near chemical storage, someone in the water, a gas leak, or an electrical arc. Press the E-stop button on the main panel (red, mushroom-head button). This cuts power to all process equipment simultaneously.

**Warning:** E-stop does NOT close valves — water may continue to flow by gravity. After pressing E-stop, manually close the intake and distribution valves if it is safe to do so.

After an E-stop: call emergency services if needed, then notify your supervisor. Do not restart the plant after an E-stop without supervisor authorization and a full inspection.

### Extended Shutdown (More Than 24 Hours)

1. Drain and flush chemical dosing lines to prevent crystallization — especially alum and phosphoric acid lines.
2. Leave filter media wet. Drying out damages the bed structure and biological layer.
3. Maintain at least a minimal chlorine residual in the clear well to prevent bacterial regrowth.
4. Set up a daily monitoring schedule, even during extended shutdowns.
5. If temperatures will drop below freezing, drain any lines or instruments that could freeze and crack.

## Post-Shutdown Checklist

Before leaving the plant after any shutdown, confirm:

1. Logbook entry is complete — time, reason, final readings, who was notified.
2. All relevant LOTO is applied and documented.
3. Chemical rooms are secured.
4. All exterior doors are locked.
5. Supervisors and relevant contacts are notified.
6. Clear well level is recorded.
7. Any active alarms are acknowledged and documented.

## SECTION 4

# Sampling & Lab Testing

### Why Sampling Matters

Your samples are the evidence that the plant is working. A bad sample gives bad data, and bad data leads to bad decisions. Sample and test correctly every time — it protects public health and protects you legally.

**Five rules, no exceptions:** (1) Take samples the same way, same spot, every time. (2) Label the container before you fill it — location, date, time, initials. (3) Test fast — most parameters (especially chlorine) degrade within minutes. (4) Write down the result immediately, don't rely on memory. (5) Never alter a result. If a number looks wrong, re-sample to confirm — falsifying records is a serious offence.

### Collecting a Grab Sample

A grab sample is a single snapshot in time. It's the right method for turbidity, pH, temperature, chlorine, and most routine parameters.

1. Run the tap for 2–3 minutes before collecting, to flush stagnant water out of the line.
2. Hold the bottle downstream of the flow — not directly under the tap.
3. Fill to the required volume. Don't fill to the brim if the bottle has a preservative in it.
4. Cap immediately.
5. Label with location, date, time, and your initials.
6. Test it within the holding time for that parameter (see the container/holding-time table below).

**Exception:** Lead and copper samples must NOT be flushed first — check your protocol before sampling for these.

### Sample Points and What to Test

This is the standard set of checkpoints through the plant, from raw water to the point it leaves for distribution:

- **Raw water intake:** turbidity, pH, temperature every shift; alkalinity daily (alkalinity sets your coagulant dose).
- **Settled water (clarifier effluent):** turbidity and pH every 2–4 hours — should be noticeably lower than raw water.
- **Filter inlet:** turbidity every 2–4 hours — should match the settled water reading.
- **Filter effluent:** turbidity continuously online, plus one manual grab per shift. This is a key compliance parameter.
- **Finished water (entry to distribution):** turbidity, free chlorine, total chlorine, pH, and temperature every 2–4 hours (minimum once per shift). This is your primary compliance point.
- **Clear well:** free and total chlorine every 4 hours, to verify residual for CT calculation.
- **Distribution system (remote points):** free chlorine and pH per your sampling plan, to confirm residual holds to the end of the system.
- **Backwash waste:** turbidity during backwash, to confirm when the backwash is complete.

**This site's compliance check point is the bag filter outlet / carbon vessel outlet — turbidity by NTU meter. The sand filter is bypassed here, so turbidity is NOT checked at a sand filter effluent.**

## Testing Turbidity (NTU)

Use a bench-top or portable turbidimeter (e.g. Hach 2100Q, Hach TU5).

1. Rinse the sample cell twice with sample water and discard both rinses.
2. Fill the cell to the fill line — don't overfill.
3. Wipe the outside of the cell with a lint-free cloth (fingerprints throw off the reading).
4. Place it in the instrument, cap it, and press READ.
5. Record the result in NTU.

**Watch for:** air bubbles give a falsely high reading (tap the cell gently to release them); scratched cells scatter light and should be replaced; an uncalibrated instrument should be calibrated before the shift per its manual.

**Testing pH** — use a bench-top or portable pH meter. **Rinse the probe tip with clean/distilled water immediately before every single test** — before each calibration buffer and before every sample, no exceptions. Skipping this carries residue from the last reading into the next one and throws off the result.

1. Calibrate the electrode with pH 7.0 and pH 4.0 (or 10.0) buffer solutions before each shift, rinsing the electrode between each buffer.
2. Rinse the electrode with distilled water. Blot dry — do not rub.
3. Immerse in the sample and wait for the reading to stabilize (typically 30–60 seconds).
4. Record the result.
5. Rinse with distilled water before the next sample, if any.
6. When finished, rinse with distilled water and store the probe in electrode storage solution — never in plain distilled water, which damages the electrode.

**pH affects almost everything downstream: coagulant works best in a specific pH range, chlorine disinfects far more effectively at lower pH, and pH that's too high or too low corrodes pipes.**

## Testing Free and Total Chlorine (mg/L)

Method: DPD colorimetric — DPD-1 measures free chlorine, DPD-3 measures total chlorine.

1. Collect the sample directly into the test vial. Don't let it sit — chlorine degrades within minutes.
2. Fill to the correct volume (usually 10 mL).
3. Add one DPD-1 tablet (or powder pillow) for free chlorine. Cap and invert gently 3 times to mix.
4. Within 60 seconds, compare the colour to the comparator wheel or read it in a colorimeter. Record the free chlorine result.
5. For total chlorine, add a DPD-3 tablet to the same vial and read again within 60 seconds. Record the total chlorine result.
6. Combined chlorine = total chlorine minus free chlorine.

**Free chlorine is the active disinfectant. If free chlorine is very low but total chlorine is high, you likely have a chloramine (combined chlorine) problem.**

**Watch for:** waiting too long to read gives a falsely low result (chlorine dissipates fast); old or damp DPD reagent won't develop colour properly; very high chlorine bleaches the indicator and gives a falsely low reading — dilute the sample 1:10 and multiply the result by 10.

## Testing Temperature and Alkalinity

**Temperature:** immerse a thermometer or probe in the sample (or in-situ at the sample point), wait for the reading to stabilize, and record in °C. Temperature affects chlorine demand (colder water needs more contact time), coagulant dose (cold water coagulates poorly), and biological activity.

**Alkalinity:** measured by titration with sulphuric acid, reported as mg/L as  $\text{CaCO}_3$ . Alkalinity is the water's ability to buffer pH changes and is critical for coagulation — alum consumes alkalinity, so low-alkalinity water may need a supplement (lime or sodium bicarbonate) added to allow coagulation to work properly.

## Jar Testing for Coagulant Dose

A jar test is a miniature, bench-top version of full-scale coagulation/flocculation. Run it to find the optimal coagulant dose before changing anything on the full-scale process. It takes 30–40 minutes and saves hours of poor treatment downstream.

1. Fill six identical 1-L beakers with raw water.
2. Set up the jar test apparatus (gang stirrer) with all six paddles in place.
3. Dose each beaker with a different coagulant amount (e.g. 5, 10, 15, 20, 25, 30 mg/L).
4. **Rapid mix:** stir at high speed (100–150 RPM) for 1 minute to simulate rapid mixing.
5. **Slow mix:** reduce to low speed (20–30 RPM) for 15–20 minutes and watch floc form.
6. **Settle:** stop the stirrers and let settle for 10–15 minutes.
7. Carefully draw off settled water from each jar without disturbing the floc on the bottom.
8. Measure turbidity and pH of each sample.
9. The jar with the lowest settled turbidity at the correct pH is your optimal dose — adjust the full-scale dose to match.

### Sample Containers, Preservation, and Holding Times

- **Turbidity:** plastic or dark glass, no preservative, analyze within 24 hours (sooner is better).
- **pH:** plastic or glass, no preservative, analyze immediately.
- **Free/total chlorine:** plastic, no preservative, analyze immediately.
- **Temperature:** measured in-situ — no container needed.
- **Alkalinity:** plastic, kept on ice (4°C), analyze within 24 hours.
- **Coliform bacteria:** sterile plastic bottle pre-dosed with sodium thiosulfate, kept on ice (4°C), analyze within 6–24 hours (confirm with your lab).
- **Metals (iron, manganese):** plastic, preserved to pH < 2 with nitric acid, holding time 28 days.

**Never** re-use a sample container for bacteriological testing without re-sterilizing it first.

### Quality Assurance Checks

- Calibrate instruments before each shift and log the calibration check results.
- Periodically run a blank (distilled water) and a known-concentration standard. If the standard reads outside  $\pm 5\%$ , the instrument needs service.
- Occasionally take duplicate samples from the same point at the same time and analyze both — results should agree within  $\pm 10\%$ .

## SECTION 5

# Chemical Dosing

### 5.1 The Chemicals On This Site — Quick Reference

This is an iron-removal and hydrocarbon-removal system, not a drinking water plant. Never mix chemicals together — several combinations on this site are dangerous (see 5.2).

| Chemical                                         | Container                                           | Job                                                                                                                                                                  |
|--------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PACI</b> (Poly Aluminum Chloride)             | White 200L drum, Class 8, SUMAS label               | Coagulant — neutralizes particle charge so solids clump                                                                                                              |
| <b>CL-200</b> (polymer flocculant)               | White drum, "FLOCCULANT" label                      | Flocculant — grows small clumps into big, heavy floc. Added at 2 confirmed points: the oil/water separation cell ("the lagoon") and the C Can                        |
| <b>Flopham</b> (dry polymer)                     | White bags, "NON-REGULATED" label                   | Backup/supplemental flocculant — same role as CL-200                                                                                                                 |
| <b>HaloKlear LiquiFloc 2%</b> (chitosan acetate) | —                                                   | Natural chitosan flocculant — used for fine silts, especially stormwater                                                                                             |
| <b>Lime — quicklime (CaO)</b>                    | 50 lb bags, mixed 1 bag/200L in a tank at the C Can | Added at the C Can before PACI + flocculant — raises pH to 8+ for coagulation, also precipitates dissolved metal ions. Conditional on pH: only used if pH is too low |
| <b>Sodium Hydroxide (NaOH)</b>                   | Big tote bag (location TBD)                         | Alternative pH raiser to lime — present on site, not yet confirmed in routine use                                                                                    |
| <b>Hydrogen Peroxide</b>                         | Blue 200L drums, Class 5.2/8, SUMAS label           | Oxidizer — converts dissolved iron (invisible) to iron hydroxide (filterable)                                                                                        |
| <b>CO<sub>2</sub></b>                            | Air Liquide gas cylinder                            | pH reducer — lowers pH when water is too alkaline                                                                                                                    |
| <b>pH+ powder</b> (Summer Smiles Sani Marc)      | 8 kg pail                                           | pH raiser — raises pH when water is too acidic                                                                                                                       |
| <b>Accu-Tab Blue</b> (Cal-Hypo tablets)          | White pails, Class 5.1, SUMAS label                 | Disinfectant — final chlorination before discharge                                                                                                                   |

#### 5.1a pH Drives Coagulant/Flocculant Dosing

PACI coagulant and flocculant dosing are not "set and forget" — as the water's pH changes, dosing has to be adjusted in response.

**Target: pH 8 or higher is good for coagulant. Below 8 is not.**

Lime brings pH up to 8+ before PACI coagulant and flocculant go in. PACI itself is acidic and lowers pH on its own, which is part of why pH is re-checked continuously rather than dosed on a fixed setting.

*Why:* PACI works by neutralizing the electrical charge on suspended particles, and that reaction is pH-dependent — outside its optimal pH window it works poorly even at the correct dose. If pH drifts below 8, the coagulant (and often the flocculant right after it) needs to be re-tuned, not left as-is.

There is no fixed daily target dose (mg/L) for PACI or flocculant. Per the supervisor: "the dose is going to change daily depending what they dump. So you're just going to have to play with it and wait until you see it flocc." Raw water composition varies day to day, so dose is set by observation — watch for floc forming, not a fixed setpoint.

### 5.1b Ongoing Testing Drives Dose Adjustment

Dosing at the C Can is not a one-time step. The operator tests the water continuously and adjusts based on results:

- If results show under-dosing (poor floc formation, turbidity not dropping), add more coagulant or flocculant at the C Can into the weir tank.
- To change the dose itself, adjust the **stroke rate and/or stroke length** on the relevant dosing pump (coagulant, flocculant, or lime) — not just by adding chemical by hand.

**Rule: Coagulant (PACl) dose must never exceed the flocculant dose — flocculant  $\geq$  coagulant, always.**

**pH check during testing at the C Can:** while testing, check the pH live. Per the target range in 5.1a (8+ good, below 8 not), add lime to bring it up to 8+ if needed (see 5.6a).

Still pending: photos/specs of the coagulant and lime dosing pumps at the C Can (flocculant pump is documented below); the exact test result that triggers a dose increase vs. a pump adjustment.

### 5.2 Critical Compatibility Warning

Several chemicals on this site react violently if mixed. Never combine:

- **Hydrogen Peroxide + Calcium Hypochlorite (Cal-Hypo)** — violent decomposition, fire, chlorine gas release
- **Hydrogen Peroxide + any organic material** (rags, wood, oils) — can ignite spontaneously
- **Cal-Hypo + any acid** (including PACl, pH 2.2–2.8) — releases chlorine gas
- **CO<sub>2</sub> gas + enclosed, unventilated space** — displaces oxygen, asphyxiation risk

Store each chemical in its designated location. Never move containers without supervisor approval. Never decant into an unlabelled container. Never combine leftover chemicals.

### 5.3 PACl (Poly Aluminum Chloride) — Coagulant

**Product:** CK-311 Coagulant (Aluminum Hydroxide Chloride Solution), SUMAS Environmental Services. SDS supplier: CARBONeT Holdings Inc., 604-761-2308.

**Container:** White 200L drum, Class 8, UN 3264, PG III, pH 2.2–2.8.

**Pump:** Blue diaphragm dosing pump.

Water contains tiny negatively-charged particles too small to settle on their own — they repel each other like tiny magnets. PACl is positively charged; when added and mixed, it neutralizes that negative charge so particles can stick together (**coagulation**). Without it, fine particles pass straight through every filter — coagulation is always the first chemical step.

**Typical dose: 5–40 mg/L (highly variable — start with baseline, adjust by observation)**

Dose point: added to water with good mixing (turbulent/rapid mix). Check the pump is running and the tank level is visibly dropping over a shift.

**Preparing the dry PACl stock solution** (25kg bags, mixed separately from the liquid CK-311 drums):

1. Mix **1 bag of dry PACl powder per 100L of clean water** — not the water being treated.
2. **Let it sit 3 hours before using.** Don't dose straight from a fresh batch — plan ahead so stock doesn't run out mid-shift.
3. Mix in the tank, then meter the resulting solution into the process via the dosing pump.
4. Indoors, coagulant and flocculant are injected together right before the water reaches the white settling tanks — the combination clumps contaminants and dirt so they settle.
5. Adjust the dosing pump's stroke rate/length as needed — watch visually for good floc formation as your guide.

**Signs of under-dosing:** tiny pinpoint particles that don't clump; water stays turbid after settling; filters get dirty quickly.

**Signs of over-dosing:** turbidity gets worse (charge reversal — particles go positive and repel again); unusually high aluminum in discharge water; pH drops sharply.

Corrosive (Class 8) — causes skin and eye burns. PPE: chemical splash goggles, chemical-resistant gloves, apron. Eye contact: flush at eyewash for 15 minutes.

## 5.4 CL-200 Polymer Flocculant

**Product:** CL-200 Water Treatment Flocculant, aqueous solution of water-soluble polymer.

**Container:** White drum, "FLOCCULANT / NON-REGULATED MATERIAL" label.

**Pump (indoor injection point):** Yellow LMI diaphragm dosing pump.

**Pump at the C Can:** Blue **JCMB Solenoid Dosing Pump**, Model JCMB55-20/2.5, capacity 20 L/H, pressure 2.5 bar, power 30W, AC100–240V.

**Current setting: flocculant pump speed adjusted to 48.0**

Coagulation (PACl) creates small particle clusters. Flocculation grows those into larger, heavier floc that settles faster and filters more easily. CL-200 is a long-chain polymer that acts like a net, bridging coagulated particles into larger aggregates. Coagulation alone produces fine, slow-settling "pinpoint floc" — adding polymer turns hours of waiting into minutes.

**Two confirmed injection points** (an earlier note describing four points, including "the lagoon" as a separate fourth point, was incorrect — "the lagoon" is just another name for point 1; the operator's job ends once treated water reaches the storage tank, with no dosing step after that):

1. **Oil/Water Separation Cell** (outdoor, Step 2 — also called "the lagoon") — added early to start clumping fine particles before water reaches the frac tanks.
2. **The C Can** — dosed alongside PACl coagulant, right before the outdoor weir tank.

Not yet confirmed: older notes mention "two Blue Separator Bins" and a "peroxide oxidizer" dosing point for flocculant — these haven't been confirmed as part of the current live process.

*Why multiple points:* each point treats water that has already lost some solids but picked up disturbance from pumping. Re-dosing at each major transfer point keeps floc forming continuously — like reapplying glue at each stage of an assembly line rather than once at the end.

General rule: dose after PACl has had time to work, during gentle (not violent) mixing, so the polymer can bridge particles without tearing floc apart. Typical dose: 0.1–2 mg/L per injection point. Over-dosing polymer is easy and wastes money — start conservative.

**Preparing the flocculant stock solution** — ratio: **1 kg polymer per 1000L water (1 g/L)**, added in two half-volume pours:

1. Add the first half of the water (not the water being treated) to the tote.
2. Turn on the mixer.
3. Add the polymer while adding the second half of the water at the same time — pouring water and polymer together while mixing lets the polymer disperse instead of clumping.
4. Keep mixing until fully dissolved.
5. **Let it sit 24 hours before use.** Plan batches a day ahead so you don't run out mid-shift.

Confirmed batch size: **500L**, scaled to the same ratio — 250L water → mixer on → 500g polymer added with the second 250L of water.

Discrepancy not yet reconciled: an older account described a 200L batch at roughly 100g polymer per 200L (~500 mg/L) — about half the concentration of the current 1 kg/1000L (1000 mg/L) ratio above. Use the current 1kg-per-1000L instructions as the live procedure; the older ratio is kept only for reference.

This stock is metered into the process alongside PACl coagulant — together they clump contaminants and dirt so they settle.

**Signs of under-dosing:** floc forms but stays small and slow to settle; settled water still turbid.

**Signs of over-dosing:** floc may break apart or get sticky ("restabilization"); water can become viscous or stringy.

Non-regulated, pH 6.5–7.5, low hazard — but can make floors very slippery if spilled; clean up immediately. PPE: gloves recommended.

### 5.5 Flopham Dry Polymer (Backup Flocculant)

**Product:** Flopham water-soluble polymer (polyacrylamide-based).

**Container:** White bags, "NON-REGULATED MATERIAL" SUMAS label.

Same job as CL-200 — used as a backup or supplement when CL-200 is unavailable or higher doses are needed. Dry polymer must **never** be added directly to the treatment stream — it forms fish-eye lumps and clogs lines.

1. Prepare a mixing bucket with clean, room-temperature water (cold water slows dissolution).
2. Slowly add dry polymer powder while stirring continuously — add powder to water, not water to powder.
3. Stir 30–60 minutes until fully dissolved and clear/uniform.
4. Typical concentration: 0.1–0.5% (1–5 g polymer per litre).
5. Add the prepared solution to the treatment stream via dosing pump or measured addition.

**Why:** pre-dissolving eliminates clumping and ensures even distribution — undissolved lumps pass through the system without doing anything useful.

### 5.6 HaloKlear LiquiFloc 2% (Chitosan Flocculant)

**Product:** HaloKlear LiquiFloc 2%, HaloSource Inc., Bothell WA.

**Chemical:** Chitosan acetate solution — a natural biopolymer derived from crustacean shells.

**Appearance:** Clear to pale yellow viscous liquid, vinegar odor.

A natural, biodegradable flocculant particularly effective for removing fine clay and silt from stormwater — preferred in some applications for being non-toxic and biodegradable.

**When to use:** high-turbidity events (heavy rain, surface runoff); stormwater with fine clay that PACl alone doesn't handle well; environmentally sensitive applications needing a natural flocculant.

**Typical dose: 1–10 mg/L (100% biodegradable, nontoxic to aquatic life)**

Very low hazard, may mildly irritate eyes. Vinegar smell is normal. PPE: gloves recommended as good practice.

## 5.6a Lime — pH Raiser / Metals Precipitant

This is the "3rd chemical" at the C Can.

**Chemical:** the bag label reads "TEXLIME High Calcium Hydrated Lime," but the supervisor (Brendan), asked directly, said: *"So it's quicklime which is calcium oxide, the bag we should around and it just says lime."* Treat this as **quicklime (CaO)**, not hydrated lime ( $\text{Ca}(\text{OH})_2$ ) — the supervisor's word overrides the label.

**Ratio: 1 bag (50 lbs) per 200L of water**

Must be kept mixing continuously — it will settle out and harden if mixing stops. Mixed dry-bag-into-stock-tank, same pattern as PACI/flocculant; the blue drum/tote staged at the C Can is this mixing tank.

**Critical:** the mixer blade on the blue lime drum must be running while you're pumping lime out, not just while first mixing — lime settles out fast, and if the blade stops while dosing, it won't work well. This is the pneumatic (air-driven) tote mixer (Section 6.16) — open the air valve and keep it running the whole time lime is being drawn out. Photo-illustrated quick steps for mixing lime, PACI, and flocculant (and running the mixer) are in [CHEMICAL\\_MIXING\\_GUIDE.pdf](#).

**Alternative chemical:** per the supervisor, sodium hydroxide can also be used — "they have a big tote bag worth of" it. NaOH is a separate, more hazardous strong base that can substitute for lime as a pH-raiser; present on site but not yet confirmed as in current routine use (see 5.6b).

Supervisor's caution, applies to both lime and NaOH: "Both raise pH so don't add too much."

**Dosing logic at the C Can:** per the supervisor — *"Lime you only need to use if the pH is too low. If it's too low sometimes the chemicals won't work."* Lime is conditional on the pH reading, not pumped on a fixed every-batch schedule. (Earlier conflicting statements — one matching this conditional logic, one saying "we pump the lime every time because the coagulant lowers the pH" — are reconciled this way: check pH every time, add lime whenever the reading is below 8. In practice this can look like "every time" if pH is consistently low, but the decision point is the reading, not the clock.)

**Working threshold: pH 8 is the hard cutoff — anything lower needs lime**

**What it does:** PACI works well at pH 8+ and is itself acidic, so without lime to counteract it, the coagulant would drag pH below its own optimal range.

**Second function:** per the supervisor — *"If you add the lime it will force any metal ions dissolved to transform into solid particles that can be filtered out in the treatment process if you don't trust the ozone."* Raising pH with lime can push dissolved metal ions (like iron) past their solubility limit so they precipitate as solid metal hydroxide particles the filters can catch — a separate mechanism from pH-correction-for-coagulant, achieving a similar end result to ozone/peroxide oxidation (Section 5.7) but via a pH shift rather than oxidation. This appears to be a backup/parallel pathway for metals removal, not a replacement for ozone.

This is quicklime (CaO) — caustic/corrosive, can burn skin and eyes, dust is a respiratory irritant. Quicklime reacts exothermically (releases real heat) when wetted — a bigger handling hazard than ordinary hydrated lime. PPE: chemical splash goggles, chemical-resistant gloves, dust mask. Never add water to a sealed/closed container of it.

Still needed: PPE specifics from the SDS; where lime is stored/staged at the C Can.

## 5.6b Sodium Hydroxide (NaOH, Caustic Soda) — Not Yet Fully Documented

On site as an alternative pH-raiser to lime — a big tote bag's worth, per the supervisor. Not yet confirmed whether it's in current routine use or just available as backup.

**Chemical:** a strong, fast-acting base, more hazardous to handle than lime. Highly corrosive to skin, eyes, and respiratory tract — more aggressively so than lime. Dissolving it in water is exothermic (releases heat). Standard handling requires chemical splash goggles, face shield, chemical-resistant gloves and apron, and care to avoid generating heat too fast in a closed container.

Like lime, raises pH — don't add too much.

Still needed: exact product/concentration, container type and location, current dose/ratio if actually in use, SDS, PPE confirmation.

## 5.7 Hydrogen Peroxide (H<sub>2</sub>O<sub>2</sub>) — Iron Oxidizer

**Product:** Hydrogen Peroxide, All Grades, Brenntag Canada.

**Container:** Blue 200L drums, SUMAS label "ORGANIC PEROXIDE TYPE E, LIQUID," Class 5.2 + Class 8.

**Concentration:** likely 25–50% (confirm with supervisor and SDS).

Source water contains dissolved iron in its reduced form (Fe<sup>2+</sup>) — ferrous iron, which is invisible/clear in water and can't be filtered out because it's not a particle. Hydrogen peroxide oxidizes Fe<sup>2+</sup> to Fe<sup>3+</sup> (ferric iron), which then forms Fe(OH)<sub>3</sub> — iron hydroxide, a reddish-brown solid that precipitates and can be filtered. This is where the orange staining on every surface comes from.

**Why:** you cannot filter dissolved iron — it must be converted to a particle first. H<sub>2</sub>O<sub>2</sub> does that conversion; the sand filter catches the resulting particles.

**Rule of thumb: 1 mg H<sub>2</sub>O<sub>2</sub> per mg of Fe<sup>2+</sup> (1:1 ratio); typically 1–3 mg/L H<sub>2</sub>O<sub>2</sub> in practice**

Excess H<sub>2</sub>O<sub>2</sub> is not harmful in small amounts — it decomposes to water and oxygen.

**Checking it's working:** orange/rusty precipitate forming in settling tanks is a good sign. If no precipitate forms, check the pump is running, tank level is dropping, and pH is in range (oxidation works best at pH 5–8).

Class 5.2 Organic Peroxide and Class 8 Corrosive — both apply. Concentrated H<sub>2</sub>O<sub>2</sub> causes severe skin/eye burns. Strong oxidizer — reacts violently with organics (rags, wood, fuels), metals (iron, copper, manganese), and strong bases. Never mix with Cal-Hypo. Decomposition produces oxygen — sealed/hot containers can over-pressurize.

**Storage:** cool, dry, away from combustibles, no direct sunlight, away from other chemicals, never near Cal-Hypo.

**PPE:** full face shield, chemical splash goggles, chemical-resistant gloves (neoprene/nitrile, check SDS), apron.

**First aid — skin:** flush with large amounts of water 20 minutes, remove contaminated clothing while flushing.

**First aid — eyes:** eyewash 15 minutes minimum, seek medical attention.

**First aid — inhalation:** move to fresh air, call 911 if breathing affected.

## 5.8 CO<sub>2</sub> (Carbon Dioxide) — pH Reducer

**Product:** Carbon Dioxide Gas/Liquid, Air Liquide Canada.

**Container:** Pressurized cylinder (high-pressure gas — handle with care).

**Emergency contact:** Air Liquide Canada, (514) 878-1667.

CO<sub>2</sub> dissolved in water forms carbonic acid (H<sub>2</sub>CO<sub>3</sub>), which lowers pH. Unlike strong acids (sulphuric acid), CO<sub>2</sub> is self-limiting — the more added, the more equilibrium pushes back — and leaves no harmful residual.

**When to use:** incoming water pH too high/alkaline (e.g. after lime or concrete contact); need to bring pH down before a step requiring a certain range.

High pressure gas — cylinders must be chained to a wall or fixed support; a falling cylinder can rupture and become a projectile. CO<sub>2</sub> gas is heavier than air — in an enclosed/low space it displaces oxygen without warning (no smell, no colour). Asphyxiation risk. Always work in ventilated areas; never enter a CO<sub>2</sub> storage vault alone.  
PPE: safety glasses, gloves. For CO<sub>2</sub> liquid contact: thermal gloves (rapid evaporation causes frostbite).

## 5.9 pH+ Powder (Summer Smiles Sani Marc) — pH Raiser

**Product:** Summer Smiles pH+ (pool-grade sodium carbonate/bicarbonate blend).

**Container:** 8 kg pail.

Sodium carbonate (Na<sub>2</sub>CO<sub>3</sub>) is a mild base — the same product used in swimming pools to raise pH.

**When to use:** water pH too low/acidic (e.g. after heavy PACl dosing, which is acidic at pH 2.2–2.8); when discharge requirements specify a minimum pH (usually ≥ 6.5).

1. Wear gloves — powder can irritate skin and eyes.
2. Add powder slowly to a turbulent area (flowing water, not a stagnant tank).
3. Add a small amount, mix thoroughly, test pH, then add more if needed.
4. Do NOT dump a large amount all at once — pH can overshoot.

**Why:** pH adjustment requires patience — a little, mix, test, repeat. Overcorrecting costs time and extra chemical to correct back.

## 5.10 Accu-Tab Blue (Calcium Hypochlorite Tablets) — Disinfectant

**Product:** Accu-Tab Blue, Calcium Hypochlorite.

**Container:** White pails, SUMAS label, UN 2880, Class 5.1 (Oxidizer).

**Concentration:** hydrated mixture, 5.5–16% water content.

Cal-Hypo dissolves in water to release hypochlorous acid (HOCl) — the same active disinfectant as liquid bleach. Tablets dissolve slowly as water passes over them in a tablet chlorinator/contact chamber. This is the final disinfection step — kills remaining bacteria before discharge and provides a measurable free chlorine residual to confirm treatment is complete.

1. Confirm the contact vessel/chlorinator is clean and dry before adding new tablets.
2. Wear gloves; do NOT inhale dust — tablets are strong oxidizers, dust irritates lungs and eyes.
3. Add tablets per dosing SOP — never to a vessel with wet residue from a previous batch (especially peroxide residue). Dry vessel is critical.
4. Allow water to flow through at the designed rate — tablets dissolve proportionally to flow.
5. Test free chlorine residual in the effluent with a DPD test kit to confirm correct dose.

Target free chlorine for discharge not yet confirmed with supervisor/permit — typical range cited is 0.1–0.5 mg/L at the discharge point.

Class 5.1 Oxidizer — can cause fire on contact with organics (wood, paper, clothing). Never mix with acids (including PACl) — releases chlorine gas. Never mix with hydrogen peroxide — violent decomposition. Store cool, dry, well-ventilated, away from all other chemicals.

PPE: chemical splash goggles, gloves, apron. If handling dusty tablets: N95 or half-face respirator.

## 5.11 Chemical Dose Calculation

All doses are proportional to the volume of water being treated. In a batch system (frac tanks):

**Dose (mg/L) × Tank volume (L) = Total chemical mass (mg); Total mass (mg) ÷ Concentration (mg/mL) = Volume to add (mL)**

**Example — adding PACl to a frac tank:** target dose 20 mg/L, frac tank volume 50,000 L (50 m<sup>3</sup>), PACl concentration ≈ 10% = 100 mg/mL.

20 mg/L × 50,000 L = 1,000,000 mg total → 1,000,000 mg ÷ 100 mg/mL = 10,000 mL = **10 L of PACl solution.**

*Why:* in a batch system you add a fixed amount to a known volume rather than running a continuous feed — getting the math right matters: too little means poor treatment, too much means wasted chemical and potential exceedances.

## 5.12 Chemical Inventory Checklist

Check at the start of every shift; record in the daily log.

| Chemical                       | Reorder Threshold | Supplier                   |
|--------------------------------|-------------------|----------------------------|
| PACl (white drum)              | < 20% remaining   | SUMAS 1-250-374-4151       |
| CL-200 polymer (white drum)    | < 20% remaining   | SUMAS 1-250-374-4151       |
| Flopham polymer (bags)         | < 2 bags          | SUMAS 1-250-374-4151       |
| Hydrogen peroxide (blue drums) | < 1 full drum     | SUMAS 1-250-374-4151       |
| Cal-Hypo tablets (white pails) | < 1 full pail     | SUMAS 1-250-374-4151       |
| CO <sub>2</sub> cylinder       | Cylinder < ¼ full | Air Liquide 1-800-817-7697 |
| pH+ powder                     | < 2 kg            | (confirm supplier)         |

## Equipment Guide

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### 6.1 Outdoor Settling Ponds

Two large concrete-block-walled open basins outdoors. The **primary settling pond** (turbid green/grey) receives raw site water first, where heavy solids settle out. The **stabilization/secondary pond** (dark red or purple, iron-rich) receives more heavily contaminated water for oil/water separation and additional settling.

**Gravity pre-settling: given hours to days, solids sink, oil floats, and clearer mid-layer water is drawn off downstream.**

Each shift, check: water level in each pond (rising or falling), any oil sheen on the surface, debris blocking the outlet pipe, and that the concrete block walls are intact.

### 6.2 Indoor Storage Tanks — Tank 1 and Tank 2

Two large white cylindrical tanks (poly/fibreglass, roughly 3–4 m tall) inside the treatment building, each with an external ladder for level checks and a ball valve at the base. These are **storage** tanks only — they hold water at different stages, they are not where chemical floc forms or settles (that happens in the separate baffled settling bins, Section 6.2a). Water reaching this stage has already passed the bag filter and/or carbon vessel.

Water held here is also used on-site for dust control (sprayed on roads/yard). There is no strict turbidity limit for this reuse, but aim for under 10 NTU — clear with no cloudiness — so spray nozzles don't clog.

Each shift, check: water level in each tank, that the tank status label (clean/dirty/in treatment) is current, no leaks at base valves or outlet connections, and that dosing lines are injecting into the correct tank.

**Know your tanks:** mixing water from a "clean" tank into a "dirty" tank means retreating that water from scratch. Always know which tank contains what.

### 6.1a Outdoor Blue Separator Bins (Weir Tanks)

Two large rectangular steel bins, painted blue, sitting outdoors on raised concrete ecology blocks. Each has a viewing window and a vent port. Internal angled baffle plates force water to travel a longer path, giving oil and solids more time to separate before water moves indoors. The two bins run in series (not parallel) — one gravity-feeds the other through a 4" hose.

The water path: oil/water separator cell → frac tanks (oil rises, water drawn from below) → the two blue bins → submersible intake pump → indoor Tank 1 / Tank 2.

Transfers between tanks use portable submersible pumps (blue housing) dropped directly into the source tank, with a hose run to the destination — nothing runs continuously or automatically. Once the outdoor weir tank has settled, a 2-inch submersible pump is placed in the *middle* of the tank — not near the floor (avoids sludge) and not too high (avoids the oil layer) — and feeds the intake pipe running indoors. This is also the trigger point for starting ozone treatment (see 6.7).

Settled sludge at the bottom of the weir tank is removed with the AODD diaphragm pump (Section 6.12) — the same pump used for both the outdoor and indoor weir tanks.

**Valve sequencing:** always open the intake pipe valve before starting the submersible pump that pushes weir tank water indoors, and open the storage tank valve before water reaches that stage. Starting a pump against a closed valve lets pressure build up with nowhere to go.

Each shift, check: water level in each bin, any oil sheen through the viewing window, that the submersible intake pump is running and not airlocked, and no leaks at the base or access port.

### 6.1b The "C Can" — Chemical Dosing Container

A shipping container that houses the chemical dosing step: pH check, then lime if needed, then PACl coagulant and flocculant (see Section 5.6a). Water flows in from the frac tank through a check valve, gets dosed inside the C Can, and discharges to the blue weir tank for settling.

1. Two white 200L drums side by side — flocculant and PACl coagulant.
2. One blue 200L drum — lime (TEXLIME High Calcium Hydrated Lime), stored as a liquid slurry. (Hydrogen peroxide is also stored in blue drums elsewhere on site — at the C Can specifically, the blue drum is lime. Go by location, not colour.)
3. A blue diaphragm dosing pump mounted above the drums, feeding two horizontal PVC pipe runs along the wall, with small tubing taps where chemical is injected into the flow.
4. A spill kit / eyewash station mounted on the wall.

## 6.2a Indoor Settling Bins ("Wet Cell")

Multi-compartment welded steel tanks inside the building, with baffled partitions that force water to travel a longer path so solids settle. Heavy dark staining (oil and iron) on the walls is normal — it's the visual record of treatment working, not neglect.

This is where water lands right after ozone treatment, not just after chemical dosing. Full sequence: outdoor weir tank → ozone-fed via the "down" pipe → these settling bins, to settle further → submersible pump pushes it out via the "up" pipe → bag filter → carbon vessel. Floc formed by PACl and flocculant settles to the floor here and must be periodically removed.

**No fixed settle time — it's judgment-based. Water should look as clear as tap water before moving on. Settling only really happens after chemical dosing; without chemicals in the water, it won't form floc.**

Each shift, check: water level is at normal operating level, sludge isn't accumulating faster than usual, outflow looks noticeably clearer than inflow, and no oil is floating on top.

**Sludge removal:** pump accumulated sludge from the bottom using the AODD pump (Section 6.12) to the filter press (Section 6.11) or to the stabilization pond as directed.

**Confined space:** the wet cell is a confined space. Never enter without a confined space entry permit, atmospheric testing, and a trained attendant outside — the iron-laden environment can generate hydrogen gas and deplete oxygen.

## 6.3 Chemical Dosing Room

Houses the drums and metering pumps for each treatment chemical. Each pump draws from its drum and injects chemical into the water stream at the correct dose.

1. **PACl (coagulant):** white 200L SUMAS drum, Class 8 label, blue diaphragm metering pump.
2. **CL-200 (flocculant):** white drum labelled "NON-REGULATED MATERIAL — FLOCCULANT," yellow LMI diaphragm dosing pump, injected after PACl, before/during slow mixing in the wet cell.

Confirmed physical path: dosing drums → wall-mounted PVC manifold → a branch pipe drops straight down into the settling tank (Section 6.2a).

**Checking a dosing pump is working:** listen for a rhythmic clicking/ticking (diaphragm strokes); confirm the drum level is dropping over a shift; hold a rag near the injection point to see a small squirt/drip with each stroke. If the pump sounds right but nothing comes out, check for a cracked injection check valve or a clogged injection point.

**Priming a dry pump:** disconnect the discharge tubing (hold a rag over the end), run the pump until liquid — not air — flows continuously, then reconnect.

**PPE required:** wear chemical-resistant gloves and splash goggles when working on any dosing pump — chemical may spray from a disconnected line.

## 6.4 Sand Filter (SRS-SF-11) — Not Currently Used

This site does not use the sand filter. Water flows straight from settling to the bag filter and carbon vessel only — that is the normal process here, not a temporary workaround. Because this step is skipped, the bag filter takes on the full solids load, so watch it closely and change it more often than you would otherwise.

Two green cylindrical pressure vessels stand side by side, connected by a shared pipe manifold — a duplex (twin) arrangement. A separate backwash control box (Alex-Tronix, Model F2AC/DC-D) automates the backwash cycle when the filter is in use, but the most likely fault is the grey solenoid valve actuators, not the timer board itself. Since the sand filter isn't used at this site, repairing it is not an active task.

## 6.5 Carbon Vessel (SRS-CV-15)

Activated carbon adsorption. Granular activated carbon binds dissolved petroleum hydrocarbons, colour compounds, and odour-causing organics as water passes through — the only way to remove dissolved hydrocarbons, since they pass straight through sand filtration. This vessel is the second filtration stage, right after the bag filter, since the sand filter isn't used.

**Current condition:** this vessel has sludge, buildup, and old/spent carbon inside. It must be cleaned out before starting any water treatment — this is a one-time clean-out, separate from routine maintenance below.

Each shift, check: water is flowing through with no unusual pressure, outlet water has no petroleum smell, effluent has no colour, and check turbidity at this outlet with the NTU meter.

**Check for exhaustion every 1.5 months.** Signs the carbon needs replacing: petroleum odour returns, or brown/yellow tint appears in the effluent.

**Backwashing does not restore capacity:** once carbon is chemically saturated, no amount of backwashing brings it back — it must be physically replaced. Exhausted carbon is regulated waste, disposed of through an approved contractor.

**Backwash weekly** — this is routine bed maintenance (breaks up channeling, flushes trapped fines), separate from the 1.5-month exhaustion check. Procedure: close inlet and outlet valves, open the backwash drain valve, open the backwash supply valve to lift and expand the bed, run until discharge is clear of fines, then close backwash valves and reopen inlet/outlet. Record time and duration in the log.

## 6.6 Bag Filter Housing

A blue/dark cylindrical housing about 1 metre tall. Les Hall Filter brand bags: 100-micron pre-filter bags catch coarse particles, 10-micron polishing bags remove fine particles. This is the **first** filtration step the water hits (ahead of the carbon vessel), since the sand filter isn't used — it's carrying the full solids load alone, so check and change bags more often.

### To change a bag:

1. Close the inlet and outlet valves on the housing.
2. Open the pressure relief/vent valve to release pressure — have a bucket ready, there will be water inside.
3. Undo the top closure ring and lift the lid.
4. Pull the old bag out by the collar over a bucket; it will drip.
5. Inspect the old bag — note solids loading and colour (iron vs. hydrocarbon) in the logbook, then discard it.
6. Install the new bag, making sure the collar seats fully — a poorly seated collar lets water bypass the bag entirely.
7. Replace the lid, tighten the closure ring, slowly open the inlet first then the outlet, and check for leaks at the top.

**Check the bag filter every 1–2 weeks.** Check sooner if flow rate drops noticeably, after any major rain event or wet cell cleanout, and check turbidity here with the NTU meter.

## 6.7 Ozone System (AERQUS + CD30TIX)

Ozone is the first thing that happens once water leaves the weir tank and enters the building — not a polishing step at the end. The CD30TIX generator makes ozone (10% gas concentration) from an oxygen feed supplied by the AERQUS, an on-site oxygen generator/concentrator (it is not a diffuser). Running pure oxygen instead of ambient air roughly doubles the ozone concentration produced for the same unit.

The AERQUS flow meter is a rotameter, scale 0–20 SCFH, with a knob to adjust oxygen feed. If the float ball is moving and reads above 0, oxygen is actively feeding the generator and the system is working. If the ball is stuck at 0 or not moving, that indicates a blockage to troubleshoot.

**Startup order:** turn the ozone generator on first, then start the weir tank submersible pump/intake — never the reverse. Starting the pump first sends untreated raw water into the intake before ozone is actually working on it.

Sequence after ozone: the "down" pipe carries weir-tank water, ozonates it along the way, and delivers it into the indoor settling bins (Section 6.2a) to settle further. Once settled, the "up" pipe carries it on to the bag filter and carbon vessel. Small variations in ozone output aren't a crisis — the carbon vessel downstream is the deliberate backstop for whatever ozone doesn't fully oxidize. The main focus is getting the water clear.

Ozone breaks down organic contaminants (hydrocarbons), oxidizes dissolved iron into filterable particles, and disinfects by killing bacteria and pathogens.

Each shift, check: CD30TIX power indicator light on, AERQUS float ball reads above 0 SCFH, a faint ozone smell in the room (this is normal), and no alarm lights on either unit.

**Ozone is toxic to breathe:** never smell it directly or put your face near the unit. The room must be well-ventilated. Canada's workplace exposure limit is 0.1 ppm (8-hour average) — very low. If you smell strong ozone, ventilate and stand back. Ozone also degrades rubber — inspect hoses and gaskets near the system regularly.

## 6.8 Chlorination (Accu-Tab Blue Cal-Hypo Tablets)

Final disinfection step before discharge. Accu-Tab Blue Calcium Hypochlorite tablets dissolve slowly in a tablet chlorinator/contact vessel as water flows over them, releasing free chlorine. The target is a measurable free chlorine residual at the discharge point.

**To add tablets:** wear chemical splash goggles and chemical-resistant gloves; confirm the contact vessel is free of other chemical residues (especially  $\text{H}_2\text{O}_2$ ); add tablets as directed (not a full bucket at once); ensure correct flow rate (too fast under-doses, too slow over-doses); test free chlorine in the effluent with a DPD test kit; record tablets added, test result, date, and time.

**Check tablet level every shift.** Tablets dissolve continuously — running out means no disinfection, a potential permit violation.

## 6.9 Oil/Water Separator (Shell Canada System)

An above-ground separator vessel plus an underground holding tank. Petroleum is less dense than water, so it rises and is skimmed off the top while sludge and heavy solids sink and cleaner water is drawn from the middle layer. Recovered petroleum and sludge are regulated wastes — drummed, labelled, and removed by a licensed contractor; they never go to the discharge point.

**Weekly inspection:** inspect the slab and drainage points for spills/leaks; dip the underground holding tank for fuel and sludge thickness; confirm liquid level is within the acceptable range (0.8–1.1 m from tank bottom); dip the above-ground separator for fuel and sludge thickness.

| Measurement                         | Action                                                            |
|-------------------------------------|-------------------------------------------------------------------|
| Underground tank sludge > 15 cm     | Arrange sludge removal                                            |
| Underground tank fuel > 2.5 cm      | Arrange skimming                                                  |
| Above-ground separator fuel > 15 cm | Skim manually; call contractor if it recurs                       |
| Liquid level outside 0.8–1.1 m      | Check pump switch is on automatic; call contractor if it persists |
| Visible petroleum bypass at outlet  | Inspect, skim, notify Shell field engineer if it recurs           |

Log every weekly inspection on the Shell Canada log sheet: date, inspector, visual results, all dip measurements, corrective actions, waste volumes removed, and contractor/manifest number.

## 6.10 Frac Tanks

Large portable above-ground steel tanks (the type used in oil-and-gas hydraulic fracturing), typically 100,000 L capacity. They store raw site water before treatment, store treated water before discharge, move water between treatment stages, and buffer variable incoming flow so you can treat in batches.

**The clean/dirty rule:** mixing clean (treated) water with dirty (untreated) water means starting treatment over. Maintain a visible label on every tank — DIRTY/UNTREATED, IN TREATMENT, or CLEAN/TREATED.

Check levels each shift and log tank ID, level, and status. Alert your supervisor if a clean tank is getting dangerously low or a dirty tank is overflowing.

## 6.11 Kontek APC5 Filter Press (Sludge Dewatering)

A filter press squeezes water out of accumulated sludge, producing filter cakes (semi-dry solids, regulated waste removed by a licensed contractor) and filtrate water (usually returned to the head of the system for re-treatment). The wet cell and settling ponds accumulate sludge continuously — it cannot just be pumped to a drain; it must be dewatered first.

**Control panel:** keyed ON/OFF switch must be in the ON position to operate; green START button starts the hydraulic pressing cycle; red RESET button stops the cycle or resets after a fault.

**Operation:** confirm plates are closed and clamped tight, position collection bins underneath, key to ON, press START, then start the sludge feed pump. The press fills from one end, chamber by chamber. Watch the filtrate outlets on each plate — flow slowing to almost nothing means the cycle is complete. Stop the feed pump, press RESET to release pressure, then slowly open the plate stack so cakes drop into the bins. Inspect each filter cloth for tears before closing up for the next batch.

**Never open the plates under feed pump pressure. Liquid and sludge will be forced out explosively. Stop the feed pump and confirm pressure is zero first.**

Two IBC-style bins collect filter cakes below the press. When full, arrange removal through a licensed waste contractor and record date, approximate volume, waste type, and contractor/manifest number.

## 6.12 AODD Pump (Air-Operated Double-Diaphragm)

A silver/aluminum air-powered pump with two flexible diaphragms that alternate to push liquid — no electric motor, it runs on compressed shop air. This is the one diaphragm pump used to pump sludge out of **both** the outdoor weir tank and the indoor weir tank/settling bins — not separate dedicated units. It can handle thick, gritty sludge that would destroy a standard centrifugal pump.

**To operate:** connect the suction hose to the inlet and discharge hose to the outlet (confirm clamps are tight); open the inlet and outlet valves; connect the air supply hose; open the air valve — the pump immediately starts cycling with a "chunk-chunk-chunk" sound; increase air pressure to reach the desired flow (up to about 100 psi max); to stop, close the air supply valve and disconnect in reverse order.

| Symptom                                  | Cause                                           | Fix                                                           |
|------------------------------------------|-------------------------------------------------|---------------------------------------------------------------|
| Pump stalls mid-stroke                   | Air pressure too low, or pilot valve worn/stuck | Increase air pressure; cycle air off/on to restart if stalled |
| Very slow, laboured pumping              | Inlet blocked (kinked hose or solids)           | Check and clear the suction line                              |
| Air blowing out of discharge, not liquid | Diaphragm failure                               | Take offline and replace the diaphragm — keep a spare on hand |
| Chattering/fast cycling, no flow         | Air-locked (air in inlet line)                  | Prime by tilting the pump or submerging the inlet             |

**Where sludge goes:** never put pumped sludge back into the Oil/Water Separation Cell — that's the start of the process, so the same contaminants would just flow through the whole system again. Sludge should follow the regulated-waste route via the filter press (Section 6.11).

### 6.13 pH Meter and Lab Setup

A portable Apera Instruments digital pH pen/tester with pH buffer solutions: 4.0 (pink), 7.0 (green), and 10.0 (blue).

**Rinse before every single test, no exceptions:** rinse the probe tip with clean water immediately before dipping it into a buffer or a sample, every time. Skipping this carries residue from the last test into the next and throws off the reading.

**Calibrate once per day/shift** — not before every test. After calibrating, test as many samples as needed that shift, just rinsing the probe between each. Recalibrate if it's a new shift, the pen has sat unused a while, or a buffer check shows it has drifted.

1. Rinse the probe with clean water and blot dry (don't rub).
2. Place in pH 7.0 buffer, press CAL, let it stabilize, confirm it reads  $7.0 \pm 0.05$ .
3. Rinse and blot dry again.
4. Place in pH 4.0 buffer, press CAL, confirm it reads  $4.0 \pm 0.05$ .
5. Rinse and blot dry — the meter is ready to use.
6. Rinse again before dipping into the actual water sample, and before every subsequent sample.
7. After use, rinse the electrode and store it in pH electrode storage solution — never distilled water, which damages the electrode.

### 6.14 Safety Equipment in the Treatment Room

| Item                           | Location                      | Purpose                                                                                 |
|--------------------------------|-------------------------------|-----------------------------------------------------------------------------------------|
| Fire extinguisher (red, ABC)   | Wall near sand filter         | General fire suppression                                                                |
| Air horn (blue compressed-air) | Wall beside fire extinguisher | Emergency alarm / evacuation signal                                                     |
| Cylinder hand truck            | Treatment room floor          | Moving gas cylinders safely — always use it, never drag or roll cylinders on their base |

Each shift, check: fire extinguisher gauge is in the green zone, the air horn is accessible and unobstructed, and its inspection tag is current (within 1 year).

**Air horn evacuation signal:** 3 short blasts means evacuate the building immediately.

A second safety station exists in the workshop/maintenance area: an eyewash station, a spill kit, and a fire extinguisher. The site has at least two separate safety stations in two different areas — learn the route to both, not just whichever is nearest to where you usually stand.

### 6.15 CO<sub>2</sub> Cylinder Handling

CO<sub>2</sub> cylinders are heavy (about 50 kg full) and top-heavy. Always use the dedicated hand truck to move one.

1. Close the cylinder valve completely and attach the protective cap.
2. Tilt the cylinder onto the hand truck and secure the chain or strap around it.
3. Tilt the hand truck back and wheel it to the new location.
4. Secure the cylinder to a wall bracket or fixed support before removing it from the hand truck.

**Never move a CO<sub>2</sub> cylinder without the hand truck.** A falling cylinder can rupture the valve and become a pressurized projectile. Cylinders must always be chained or strapped to a fixed support when not on the hand truck.

### 6.16 Pneumatic Tote Mixer

An air-driven mixer clamped to the top of an IBC tote, used to mix PACl, flocculant, and lime stock solutions. A red-handled valve controls compressed air to the motor, which turns a shaft and propeller submerged in the liquid.

**Full steps, photos, and the keep-the-blade-running rule for pumping out lime:** see the Chemical Mixing Guide.

## SECTION 7

# Preventive Maintenance

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**⚠ Generic template — not yet site-confirmed:** This maintenance schedule is generic industry-standard content adapted from typical municipal drinking-water-plant practice, not yet customized to SUMAS Environmental Services' actual equipment, intervals, or manuals. Treat it as general background only — confirm all tasks, frequencies, and any references to equipment types (e.g. UV units, sedimentation basins, clear wells) with a supervisor before relying on it, since this is an industrial non-discharge dust-control water system, not a municipal plant.

### 7.1 The PM Philosophy

**Reactive maintenance** — fix it when it breaks. High cost, unpredictable downtime, emergency repairs in bad conditions.

**Preventive maintenance (PM)** — fix it before it breaks. Predictable, cheaper, keeps you in control. This is the target standard for most plants.

**Predictive maintenance** — monitor equipment condition (vibration analysis, oil sampling, thermal imaging) and intervene when data shows degradation. Gold standard, but needs specialized tools.

**\$50 part + 30 min now vs. thousands of \$ + plant risk later**

## 7.2 PM Frequencies and Tasks

Generic comprehensive schedule below — to be refined per piece of equipment once site equipment manuals, manufacturer-specific intervals, and maintenance records are available.

### Daily (every shift):

1. **All pumps — visual inspection:** check for unusual noise, vibration, heat, leaks. Log; escalate if abnormal.
2. **All pumps — seal/packing check:** a small drip from packing is normal, a stream is not. Adjust packing gland; schedule seal replacement if needed.
3. **Pump bearings — lubrication check:** check temperature by touch and oil level indicator if present. If hot, stop pump and check oil level.
4. **Dosing tanks — chemical level check:** record levels against reorder trigger; order if at reorder level.
5. **Turbidimeters/Cl<sub>2</sub> analyzers — instrument vs. grab sample:** must be within  $\pm 10\%$  of manual test; calibrate instrument if not.
6. **Online turbidimeters — clean flow-through cells:** remove algae/fouling with brush and DI water.

### Weekly:

1. **Pumps, mixers — lubricate bearings:** grease per manufacturer spec; do not over-grease (damages seals).
2. **All turbidimeters — calibrate:** use StablCal or Formazin standard; document.
3. **All chlorine analyzers — calibrate:** DPD manual test on same stream; adjust if needed.
4. **All pH meters/sensors — calibrate:** 2-point calibration with pH 7 and pH 4 (or 10) buffers.
5. **Peristaltic dosing pump tubing — inspect:** check for cracks, kinks, soft spots; replace before failure.
6. **Chemical injection points — check:** confirm clear, no crystal buildup, visibly injecting.
7. **Manual valves (regular use) — check packing:** adjust gland if leaking; do not over-tighten or valve becomes hard to operate.
8. **Online turbidimeter — test high-turbidity alarm:** verify trigger at setpoint and notification path.
9. **Chlorine analyzer — test low-chlorine alarm:** verify trigger at setpoint.
10. **All filters — inspect media surface:** should be even, no "rat holes" or cracking (these bypass water around the media).

### Monthly:

1. **All motors — insulation check:** use megohmmeter; record reading and compare to baseline.
2. **Motor/pump couplings — alignment check:** misalignment causes vibration and premature bearing failure.
3. **Control panels, MCC — inspect all electrical connections:** look for discolouration, corrosion, loose terminals. Do not touch live circuits.
4. **SCADA/panel — test all alarms end-to-end:** simulate fault conditions; verify alarms reach intended recipients.
5. **All chemical rooms — inspect storage area:** check for spills, container integrity, current SDS, working ventilation.
6. **Sedimentation basins — sludge blanket measurement:** record depth, compare to previous month, adjust drain schedule if rising.
7. **All filters — media inspection:** check for media loss, compaction, biological growth.
8. **Backwash pump — flow test:** confirm flow and pressure meet design spec.
9. **Clear well — inspection:** check for cracks, algae growth, animal intrusion, disinfection residual uniformity.

### Quarterly (every 3 months):

1. **Mixer/multi-stage pump gearboxes — change oil:** use manufacturer-specified oil grade.
2. **Diaphragm dosing pumps — replace diaphragms:** even if not failed, to prevent chemical spills.

3. **UV disinfection unit — inspect/clean lamps and sleeves:** wipe sleeves, measure lamp intensity, replace if below minimum.
4. **All flow meters — calibrate:** verify against a portable meter or timed fill.
5. **Normally-closed/rarely-operated valves — full exercise:** open/close fully to prevent seizing; lubricate stem.
6. **Backup generator — inspect and test:** load test at full power for 30 min; check fuel level.
7. **Chemical room ventilation fans — inspect:** confirm adequate air changes; replace filters.

**Annually (every 12 months or per permit):**

1. **All major pumps — pump curve verification:** measure flow at multiple heads vs. original curve; large deviation indicates worn impeller.
2. **All motors — winding resistance:** record and trend; significant drop signals approaching insulation failure.
3. **Pressurized vessels — pressure vessel inspection:** check internal corrosion; pressure test.
4. **All panels — comprehensive electrical inspection:** qualified electrician thermographic scan; check arc flash labeling.
5. **All instruments — calibrate against certified standards:** keep calibration certificates on file for regulators.
6. **All emergency procedures — review and update:** confirm contact numbers and equipment descriptions are current.
7. **All confined spaces (clear well, basins, wet wells) — entry permit review:** confirm procedures, equipment, and trained staff are current.
8. **All dosing pumps — overhaul:** replace all wetted parts on schedule; far cheaper than emergency repair.

### 7.3 LOTO (Lockout/Tagout) — The Non-Negotiable PM Step

Before ANY maintenance on equipment that could move, energize, or release stored energy:

1. **Identify** all energy sources (electrical, pneumatic, hydraulic, stored pressure, gravity).
2. **Notify** all affected personnel.
3. **Isolate** each energy source (breaker off, isolation valve closed, etc.).
4. **Lock** the isolation device with YOUR personal lock — each person working on the equipment applies their own lock; no shared locks.
5. **Tag** the lockout point with your name, date, and reason.
6. **Verify** zero energy state: press the start button to confirm nothing happens; check gauges for zero pressure.

Only the person who placed a lock may remove it.

**WARNING:** This is not bureaucratic paperwork. Workers are killed every year by equipment assumed to be de-energized. There are no exceptions to LOTO.

## 7.4 Maintenance Log

Every PM task should be logged with date, task, technician, findings, parts used, and next action. Example:

DATE: 2026-06-04      TASK: Weekly bearing lubrication – Raw water pumps 1, 2, 3  
TECHNICIAN: [Name]  
FINDINGS: All bearings normal temperature. RW Pump #2 – slight increase in vibration vs. last week. Greased per spec. Scheduled vibration check for next week.  
PARTS USED: None (lubrication only)  
NEXT ACTION: Monitor Pump #2 vibration.

## 7.5 Spare Parts Inventory

Maintain a minimum stock of:

1. **Dosing pump tubing (peristaltic):** fails without warning, stops dosing — keep 2 sets per pump type.
2. **Dosing pump diaphragms:** same risk — keep 2 per pump type.
3. **Pump mechanical seals:** failure causes flooding — keep 1 per pump type.
4. **UV lamp(s):** must be replaced on schedule — keep 1–2 per UV unit.
5. **UV quartz sleeve(s):** can crack — keep 1 per UV unit.
6. **DPD reagents (tablets/powder):** used daily — keep 2–4 week supply.
7. **Calibration standards (turbidity, pH):** required for daily calibration — maintain per shelf life.
8. **Chemical injection check valves:** fail from chemical exposure — keep 2–3 per injection point type.
9. **Motor contactors:** electrical failure — keep 1 per motor type.
10. **Fuses and breakers:** electrical protection — stock as per electrical panel schedule.

SECTION 8

Troubleshooting

**⚠ Generic template — not yet site-confirmed:** This troubleshooting guide is generic industry-standard content written for a conventional municipal drinking-water plant (clarifiers, filters, clear wells, distribution chlorine residual), not for SUMAS's industrial non-discharge dust-control water system. Treat it as general background only until it is reviewed and adapted with a supervisor to match the actual pond → oil/water separator → frac tanks → C Can → ozone → settling bins → bag filter → carbon vessel process.

8.1 The Troubleshooting Mindset

Troubleshooting is structured detective work: observe → form a hypothesis → test it → fix it → verify the fix → record everything.

1. **Stay calm.** Panicking makes it worse.
2. **Confirm it's real.** Could the instrument be malfunctioning? Take a manual sample and re-test.
3. **Protect the public first.** If finished water quality is in question, stop distribution and call your supervisor before anything else.
4. **Work upstream.** Most finished-water problems originate upstream — start at raw water and work forward.
5. **Change one thing at a time.** Adjusting multiple variables at once means you won't know what fixed it.
6. **Document everything.** Every observation, action, and result.

8.2 Finished Water Turbidity High

Turbidity = real-time #1 indicator of treatment performance

High turbidity at the tap means something upstream is failing. Rapid diagnosis path:

| Step | Check                                                      | If result is... | Next action                                                                                                |
|------|------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------|
| 1    | Is the turbidimeter reading accurate? (manual grab sample) | No              | Calibrate/clean instrument; monitor with manual samples                                                    |
| 1    | Is the turbidimeter reading accurate?                      | Yes             | Continue to step 2                                                                                         |
| 2    | Check filter effluent turbidity                            | High            | Filter problem — see 8.3                                                                                   |
| 2    | Check filter effluent turbidity                            | Low             | Problem is after the filters — check clear well inlet valve and for sediment disturbance in clear well     |
| 3    | Filter fine but finished water still high                  | —               | Re-check sample point; confirm clear well isn't short-circuiting; compare entry point meter vs. clear well |

### 8.3 Filter Turbidity High (Filter Breakthrough)

| Possible Cause                                         | How to Confirm                                                         | Fix                                                                              |
|--------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Filter due for backwash                                | Check headloss and run time                                            | Initiate backwash                                                                |
| Poor coagulation upstream — floc passing through media | Check settled water turbidity (low = coagulation fine; high = see 8.4) | Fix coagulation; consider backwash to reset filter                               |
| Media compaction / "mudball" formation                 | Visual inspection of media surface during backwash                     | Increase backwash intensity; air scour if available; replace media if persistent |
| Filter ran too long, exhausted adsorption capacity     | Check run time vs. normal                                              | Backwash immediately; re-evaluate max run time                                   |
| Damage to media or underdrains                         | Turbidity spike during backwash that doesn't clear                     | Take filter offline; inspect; repair                                             |
| Filter ripening after backwash                         | Elevated turbidity for first 15–30 min post-backwash, then clears      | Normal — route to waste until turbidity meets target                             |

### 8.4 Settled Water Turbidity High (Clarifier/Sedimentation Problem)

| Possible Cause                                      | How to Confirm                                                       | Fix                                                                        |
|-----------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------|
| Under-dosing of coagulant                           | Settled turbidity rising; little/no floc visible                     | Increase coagulant dose; run jar test                                      |
| Over-dosing of coagulant (charge reversal)          | High turbidity with very small floc; check dose vs. jar test optimum | Decrease dose; re-run jar test                                             |
| pH outside optimal range for coagulant              | Check post-coagulation pH vs. chemical's optimal range               | Adjust pH first, then re-optimize dose                                     |
| Raw water alkalinity too low                        | Test alkalinity after coagulant addition                             | Add lime or sodium bicarbonate                                             |
| Sludge blanket too deep — solids carrying over weir | Measure sludge blanket depth                                         | Increase sludge drain frequency                                            |
| Sudden raw water turbidity increase (storm event)   | Check raw turbidity vs. recent trend                                 | Increase coagulant dose; run jar test immediately                          |
| Mixing problem (short-circuiting in flocculation)   | Observe flow patterns; dye test if needed                            | Check weirs/baffles; ensure mixers running                                 |
| Temperature change (cold water — poor coagulation)  | Check raw water temp vs. recent weeks                                | Consider alum → PACl switch (better cold-water performance); increase dose |

## 8.5 Chlorine Residual Low in Finished Water

| Possible Cause                                                         | How to Confirm                             | Fix                                                                          |
|------------------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------|
| Chlorine dose too low                                                  | Check dosing pump output; check tank level | Increase dose; calibrate pump                                                |
| High chlorine demand in raw water (organics, iron, manganese, ammonia) | Test raw water; check NOM levels           | Increase dose to meet demand; investigate source                             |
| Chlorine dosing pump failed                                            | Visual inspection; tank level unchanged    | Switch to backup pump; repair primary                                        |
| High turbidity "hiding" chlorine demand                                | Check turbidity — fix first if high        | Fix coagulation/filtration — residual often recovers once turbidity resolved |
| Warm water increases decay rate                                        | Check temperature                          | Increase dose; adjust for seasonal conditions                                |
| Long hydraulic detention time (low flow through contact chamber)       | Check flow rate                            | Ensure minimum flow maintained; don't reduce flow too drastically            |
| Chlorine solution degraded (old/warm NaOCl)                            | Check tank: dark brown/yellow = degraded   | Order fresh chemical; store cool, dark, sealed                               |

## 8.6 pH Out of Range

| Symptom                              | Possible Cause                                                 | Fix                                                                               |
|--------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------|
| pH too low (acidic) — finished water | Coagulant consuming alkalinity; high CO <sub>2</sub> in source | Add lime or caustic; add alkalinity; reduce CO <sub>2</sub> via aeration          |
| pH too high — finished water         | Over-addition of lime or caustic                               | Reduce dose; check pH probe calibration first                                     |
| pH fluctuating erratically           | Dosing pump running inconsistently; instrument error           | Check pump for air in lines; calibrate probe                                      |
| pH drop at filter effluent vs. inlet | CO <sub>2</sub> release from biological activity in filter     | Normal if small; large drops suggest biological fouling — review filter operation |

## 8.7 Pump Problems

### Pump not starting

| Possible Cause              | Check                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------|
| Tripped breaker or overload | Check MCC/panel; reset if tripped. If it trips again immediately, do NOT keep resetting — investigate the cause |
| E-stop engaged              | Check whether any E-stop buttons are pressed in                                                                 |
| Control signal missing      | Check PLC/SCADA output; check local/remote switch position                                                      |
| Pump seized                 | Try rotating shaft manually (only when confirmed de-energized and LOTO applied)                                 |

### Pump running but no/low flow

| Possible Cause                                | Check                                                                                                            |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Discharge valve closed                        | Walk to pump and check all valves in the discharge line                                                          |
| Air lock (lost prime)                         | Check suction line for air; open vent valve                                                                      |
| Impeller clogged                              | Take pump offline (LOTO); inspect and clear obstruction                                                          |
| Pump running backwards (wrong phase sequence) | Compare rotation direction to casing arrow; if reversed, swap two electrical phases (qualified electrician only) |

### Pump vibration / noise

| Symptom               | Cause                       | Action                                                             |
|-----------------------|-----------------------------|--------------------------------------------------------------------|
| Grinding/gravel sound | Cavitation                  | Check suction pressure, reduce discharge restriction, reduce speed |
| Rhythmic thumping     | Misalignment or imbalance   | Check alignment; inspect impeller for damage                       |
| Bearing whine         | Bearing failure approaching | Schedule bearing replacement; increase monitoring frequency        |
| Loud knocking         | Water hammer                | Add slow-closing valve; check check valve                          |

## 8.8 Chemical Dosing Problems

| Problem                                       | Likely Cause                                                         | Fix                                                             |
|-----------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------|
| Dose dropping despite no pump changes         | Tank getting low                                                     | Refill; switch to backup tank                                   |
| Dosing pump running but no chemical delivered | Air in tubing; cracked tubing; failed check valve at injection point | Prime pump; inspect/replace tubing; check injection check valve |
| Dose going up but no change in treatment      | Chemical degraded (NaOCl especially)                                 | Test chemical concentration; replace if degraded                |
| Crystals forming around injection point       | Chemical concentration too high; slow flush after injection          | Flush injection line; dilute if needed; increase flush flow     |
| Peristaltic tubing failed (chemical spraying) | Tubing age / UV damage / incompatible material                       | Immediate LOTO; replace tubing; clean spill                     |

## 8.9 Alarm Response Quick Reference

| Alarm                     | Immediate Check                                | Escalate if                                                    |
|---------------------------|------------------------------------------------|----------------------------------------------------------------|
| High filter turbidity     | Manual grab sample; check upstream coagulation | Turbidity > permit limit; can't reduce within 15 min           |
| Low chlorine residual     | Manual test; check dosing pump                 | Residual below regulatory minimum; can't restore within 15 min |
| High raw water turbidity  | Check source conditions; check coagulant dose  | Turbidity > 10× normal or settled water target unreachable     |
| Pump fail / trip          | Check for E-stop, tripped breaker, overload    | After 2 reset attempts; any sign of mechanical failure         |
| High clear well level     | Check all flows; check distribution pump       | Risk of overflow                                               |
| Low clear well level      | Check all flows; check raw water pump output   | < 20% of capacity (risk of running out of water)               |
| Power failure (UPS alarm) | Confirm generator started                      | Generator doesn't start automatically within 30 sec            |

## Safety & PPE

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### The Non-Negotiable Safety Rules

**Never bypass LOTO (Lockout/Tagout):** energized equipment kills.

**Never enter a confined space alone or without a permit.**

**Always wear the required PPE.** If you don't know what's required for a task, don't start it.

If you're unsure, stop and ask — the question you feel embarrassed to ask is usually the most important one. Report every near-miss: near-misses are accidents that didn't quite happen, and they predict future accidents.

### Site Hazard Overview

1. **Chemical burns/splash** — storage rooms, dosing areas, injection points. Causes skin burns, eye damage, lung damage.
2. **Toxic gas** — chlorine room, confined spaces ( $\text{H}_2\text{S}$ ,  $\text{CO}_2$ , low  $\text{O}_2$ ). Can cause unconsciousness or death.
3. **Drowning** — open basins, wet wells, clear well.
4. **Electrocution** — MCC rooms, panels, pump controls, exposed wiring.
5. **Crushing/entanglement** — pumps, mixers, scrapers, conveyors with moving parts.
6. **Slips, trips, falls** — wet floors near pumps and basins, ladders, grates.
7. **Manual handling** — chemical drums, bags of lime, equipment. Risk of back injuries and hernias.

## PPE Guide — When to Wear What

**Handling liquid chemicals (alum, PACl, NaOCl, fluoride):** chemical splash goggles (not just safety glasses), chemical-resistant gloves (nitrile minimum; neoprene for acids), apron.

**Handling concentrated acids (fluorosilicic acid,  $\text{H}_2\text{SO}_4$ ):** face shield plus chemical splash goggles, thick acid-resistant gloves, acid apron, boots with no metal.

**Handling dry lime or powdered chemicals:** N95 dust mask, goggles, gloves.

**Chlorine room / gas chlorine systems:** SCBA (Self-Contained Breathing Apparatus) and chemical suit — trained responders only.

**Electrical work:** electrical insulating gloves (correct voltage rating) and arc flash PPE per the electrical risk assessment.

Other situations: general plant rounds need safety glasses, hard-toed boots, and a hard hat in construction zones. Working within 1 m of open water requires a life jacket and a buddy. Noise above 85 dB (large pump rooms) requires hearing protection. Working at height above 1.2 m (ladders, catwalks) requires a fall arrest harness attached to an anchor point. Entering a confined space requires the full confined space entry procedure.

### Donning and Doffing Chemical PPE

**Putting on (clean hands first):** apron, then gloves (check for holes by inflating them), then goggles or face shield.

**Removing (treat everything as contaminated):** gloves first — peel off inside-out so the contaminated surface faces in — then goggles (handle by the strap only), then apron (roll away from your body), then wash your hands immediately.

## WHMIS / GHS — Reading Hazard Labels

Every chemical container on site must have a GHS-compliant label. Know these symbols on sight:

1. Flame — flammable
2. Skull and crossbones — acutely toxic, can kill
3. Exclamation mark — irritant, less severe toxicity
4. Corrosion symbol — destroys skin and metal
5. Gas cylinder — compressed gas
6. Exploding bomb — explosive
7. Flame over circle — oxidizer, increases fire risk
8. Health hazard (starburst figure) — carcinogen, reproductive hazard, organ damage
9. Environment symbol — environmental hazard

**Safety Data Sheets (SDS):** every chemical must have an SDS on file and accessible at all times. It covers the chemical's hazards, first aid measures, firefighting measures, spill response, handling/storage requirements, and exposure limits. Know exactly where your site's SDS binder or station is.

## Lockout / Tagout (LOTO)

### The 6 steps to lock out equipment:

1. **Prepare** — identify all energy sources (electrical, pneumatic, hydraulic, gravity, stored pressure, chemical).
2. **Notify** — tell all affected workers the equipment is being locked out.
3. **Shut down** — turn off the equipment through its normal stopping procedure.
4. **Isolate** — open all disconnects, close all isolation valves, bleed stored pressure.
5. **Lock and tag** — place your own lock on each isolation point, with a tag showing your name and the date.
6. **Verify** — try to start the equipment (it should not start); check that all pressure gauges read zero.

**Removing LOTO:** confirm all workers are clear, remove all tools/materials/covers, each person removes only their own lock, notify all affected workers, then start up and verify normal operation.

## Confined Space Entry

**Never enter a confined space without a written entry permit and a trained attendant. This is non-negotiable.**

Confined spaces at a water plant include clear wells, sedimentation basins, filter underfloor access, wet wells, chemical storage vaults, and any other enclosed space with limited entry/exit not designed for continuous occupancy. Specific hazards include oxygen deficiency ( $O_2$  displaced by  $CO_2$ ,  $N_2$ , or methane), toxic gases ( $H_2S$  from biological decomposition, chlorine), flooding, and engulfment by sludge or water.

1. **Hazard assessment** — identify all hazards before opening the space.
2. **Atmospheric testing** —  $O_2$  must read 19.5–23.5%, flammable gases under 10% LEL,  $H_2S$  under 1 ppm, CO under 25 ppm for entry without supplied air. Use a calibrated gas monitor; test before entry and continuously during entry.
3. **Ventilation** — force-ventilate for at least 5 minutes before entry, and continue during entry.
4. **Entry permit** — complete and sign it, listing hazards, control measures, rescue plan, required PPE, and atmospheric test results.
5. **Attendant** — a trained, equipped attendant stays at the entry point at all times and never enters the space. Their job is to monitor and summon rescue if needed.
6. **Rescue plan** — know exactly how you would rescue an incapacitated entrant (tripod and lifeline, or other). Never attempt a non-entry rescue — jumping in to help has never gone well; you become the second victim.
7. **Communication** — maintain continuous contact between entrant and attendant.

## Electrical Safety

**Arc flash PPE:** high-voltage panels carry arc flash hazard labels. Never open one without the correct arc flash PPE for the incident energy level shown on the label.

Only qualified electricians work on live circuits. Operators should know the location of all main disconnects and emergency shutoffs. Never reset a tripped breaker more than twice without investigating the cause, and never use water near electrical equipment without de-energizing it first. Report any unusual heat, smell, discolouration, or sparking from electrical equipment immediately.

## Chemical Exposure — First Aid

Always consult the specific SDS for the chemical involved — these are general guidelines.

**Skin contact with a corrosive chemical:** flush immediately with large amounts of water for at least 20 minutes. Remove contaminated clothing while flushing. Seek medical attention.

**Eye contact with any chemical:** flush immediately at the eyewash station for at least 15 minutes. Hold eyelids open. Seek medical attention even if there's no pain.

**Inhalation of chlorine gas:** move to fresh air immediately. Call 911 if breathing is affected. Do not re-enter the contaminated area without SCBA.

**Ingestion of any chemical:** do not induce vomiting unless the SDS specifically directs it. Call Poison Control. Seek medical attention.

### Using the eyewash station:

1. Pull the activation handle to start the water flow.
2. Hold your eyelids open with your fingers — your instinct will be to squeeze them shut, override it.
3. Roll your eyes in all directions.
4. Flush for a full 15 minutes — set a timer; it feels much longer than it is.
5. Call for help — don't flush alone if you can avoid it.

## Hydrogen Peroxide Spill — Special Procedure

Hydrogen peroxide needs different spill treatment than most chemicals — **do not just flush a large spill with water.**

1. Put on full PPE: full face shield, chemical splash goggles, neoprene gloves, apron.
2. Contain the spill — keep it away from floor drains and anything combustible (wood, rags, organic material).
3. Apply a dilute sodium sulphite or sodium bisulphite solution cautiously (confirm the neutralizer is stocked on site with your supervisor).
4. The reaction will bubble and effervesce — this is expected. Stand back.
5. Flush the neutralized area thoroughly with water.
6. Dispose of waste material as hazardous waste.

**Never use sawdust, rags, or organic absorbents on concentrated  $\text{H}_2\text{O}_2$**  — fire risk. It is a strong oxidizer and will ignite organic material on contact. Flushing alone just spreads it; the reducing agent converts it into water and oxygen, which are safe.

**$\text{CO}_2$  leak in an enclosed area:**  $\text{CO}_2$  has no smell. You will feel dizziness, an irregular heartbeat, and shortness of breath before losing consciousness. Evacuate immediately, ventilate, and do not re-enter without SCBA.

### **Emergency Phone Numbers**

1. **Emergency services (fire/ambulance):** 911
2. **Poison control (Canada):** 1-800-268-9017
3. **SUMAS Environmental Services:** 1-250-374-4151
4. **Air Liquide (CO<sub>2</sub> cylinder emergency):** 1-800-817-7697

## Emergency Response

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**⚠ Partially generic template:** Some content below (distribution system entry valve, Boil Water Advisory, regulatory authority notification) reflects a municipal drinking-water plant and does not apply directly to this non-discharge industrial site (no distribution system, no discharge permit — treated water goes to on-site storage tanks for dust-control reuse only), while the general emergency response structure (chemical spill, power failure, equipment failure) is still useful as-is.

### 10.1 Universal Emergency Protocol

Regardless of the type of emergency, your first four actions are always the same:

**ASSESS → PROTECT → COMMUNICATE → DOCUMENT**

1. **ASSESS** — Is anyone in immediate danger? Are you in immediate danger?
2. **PROTECT** — Evacuate or isolate people from the hazard. If treated water quality is in question, stop sending it to the storage tank.
3. **COMMUNICATE** — Call 911 if life safety is at risk. Call your supervisor immediately. Notify SUMAS / the relevant authority as required.
4. **DOCUMENT** — Start recording from the moment you notice the emergency: time of first observation, actions taken, readings, who was called and when.

## 10.2 Chemical Spill

**Small spill** (within your training and PPE capabilities — typically < 20 L of a diluted chemical):

1. Put on full PPE (face shield, splash goggles, chemical gloves, apron) before approaching.
2. Eliminate ignition sources if the chemical is flammable (check SDS — most water treatment chemicals are not).
3. Stop the source of the spill if possible (close valve, upright container).
4. Contain the spread with absorbent berms/pigs from your spill kit, keeping it away from floor drains.
5. Absorb the spill with appropriate absorbent (check SDS — some chemicals need neutralization before disposal).
6. Collect all contaminated material in a labelled, sealed waste container.
7. Decontaminate the area with clean water.
8. Dispose of waste per your facility's chemical waste procedures.
9. Report the spill in the logbook and to your supervisor, even if it was minor.

**Large spill or any chlorine gas release** — do not attempt to control this yourself:

1. Do NOT attempt to control a large spill or any chlorine gas release yourself.
2. **Evacuate** the area immediately — upwind and uphill.
3. **Alert** all personnel on site.
4. **Call 911** — emergency responders (HAZMAT team) will manage the scene.
5. Call your supervisor.
6. Meet emergency responders at the entry point; provide the SDS and a description of what happened.
7. Do NOT re-enter until the all-clear is given by emergency responders.

Chlorine gas at even low concentrations (3 ppm) causes severe respiratory distress; at 30 ppm it is life-threatening within 30 minutes. You cannot fight it without SCBA and training — call the professionals.

### 10.3 Power Failure

#### In the first 60 seconds, check:

- Did the backup generator start automatically? (Should be audible within 15–30 seconds.)
- Are critical systems (pumps, dosing, instrumentation) on the backed-up circuit?
- Is the SCADA/HMI screen live or black?

#### If the generator auto-starts successfully:

1. Confirm all critical processes are running on generator power.
2. Check all chemical dosing pumps — some may need manual restart after a power blip.
3. Note the time of the outage in the logbook.
4. Contact the utility company to report the outage and get an estimated restoration time.
5. Monitor generator fuel level — know how long your fuel supply lasts at current load.

#### If the generator does NOT start automatically:

1. Attempt manual start at the generator (per your generator's SOP).
2. If unsuccessful, call maintenance immediately.
3. Assess how long the plant can coast on existing storage before treatment is affected.
4. Notify your supervisor (and SUMAS / the relevant authority, if required).
5. Begin load-shedding: prioritize disinfection and critical process pumps above everything else.

**Returning to grid power:** Do not switch back during a power surge — wait for stable voltage. Transfer back to grid in coordination with the generator's transfer switch procedure. After transfer, inspect all equipment for restart — some pumps and instruments may need manual reset.

### 10.4 Equipment Failure — Critical Equipment

#### Raw water pump failure:

1. Confirm the failure (trip alarm, no flow on meter, breaker tripped).
2. Start the standby raw water pump (confirm during orientation that one exists).
3. Investigate the failed pump (see Section 8 — Troubleshooting).
4. If both pumps fail: notify supervisor, manage on available storage, initiate emergency repair.

#### Filter failure (turbidity spike):

1. Take the affected filter offline (close inlet valve).
2. Divert flow to other filters (increase their load, staying within design limits).
3. Do NOT send high-turbidity filter water onward to the storage tank.
4. Backwash the failed filter; if turbidity doesn't recover, investigate (see Section 8.3).

#### Chlorine dosing failure (no chlorine residual):

1. **Immediately** stop sending water onward to the storage tank — under-disinfected water must not reach it.
2. Diagnose and fix the dosing failure (see Section 8.5).
3. Once repaired, confirm residual is restored.
4. Do NOT resume sending water forward until the finished water meets the minimum free chlorine residual.
5. Notify your supervisor / SUMAS per your site's reporting requirements.

## 10.5 Contamination Event / Adverse Water Quality

Covers: an unusual taste or odour, an unexpected turbidity spike, a microbiological positive, or a chemical detection event.

### Immediate response:

1. **Stop sending water onward** to the storage tank if the contaminant poses a risk or you are uncertain. Do not wait for confirmation.
2. **Collect and preserve samples** from all points in the treatment train immediately — these are your evidence.
3. **Notify your supervisor** immediately. If a reporting requirement applies to your site, follow it.
4. **Do not flush the system yet** — preserve the evidence first.
5. Document everything: first sign of the problem, all test results, all actions taken, all people contacted.

### Investigation — work backward through the treatment train:

- Where in the process did the contamination enter or originate?
- Was there an upstream event (spill on-site, infrastructure failure)?
- Was there a treatment failure (missed dose, equipment failure, bypassed process)?
- Was there contamination downstream of treatment (storage tank, dust-control equipment)?

**Municipal-context subsection — not applicable as written:** The original "Communication during a Boil Water Advisory" content is kept below for completeness, but it assumes a public drinking-water distribution system with a regulatory authority issuing advisories to the public. This site has no distribution system and no discharge permit — there is no public to advise. Treat this as background reference only.

### Communication during a Boil Water Advisory (original content, municipal context):

- If a Boil Water Advisory (BWA) is issued by the regulatory authority: you are a technical resource; the regulatory authority issues the advisory and manages public communication.
- Cooperate fully with the authority's investigation.
- Document every step of the corrective action.
- BWA lifting requires: corrective actions complete, adequate disinfection restored, bacteriological samples confirming safety (typically two consecutive clean samples, 24 hours apart).

## 10.6 Flooding or Extreme Weather Events

1. Monitor Environment Canada / local weather forecasts during severe weather seasons.
2. If flooding is possible: raise or sandbag critical electrical equipment; ensure the backup generator is fuelled and tested; pre-position equipment and chemical supplies; confirm contact information for all emergency services.
3. If intake turbidity rises rapidly during a storm: immediately run a jar test, increase coagulant dose, and increase monitoring frequency.
4. Extreme cold: protect all chemical lines (especially sodium hypochlorite, which freezes near 0°C), instrument sample lines, and water lines from freezing.

## 10.7 Emergency Contacts and Reporting Requirements

**Municipal/regulatory-specific language — confirm before relying on this:** The table below is the original template's reporting matrix, written for a permitted municipal plant with a distribution system and a regulatory authority. This site has no discharge permit and SUMAS Environmental Services Inc. (1-250-374-4151) is the technical supplier/contractor, not a regulator. Until site-specific reporting obligations are confirmed, treat "Regulatory authority" below as "Supervisor + SUMAS" and verify with your supervisor whether any external reporting actually applies.

| Event                                                                           | Who to Notify                           | When                            |
|---------------------------------------------------------------------------------|-----------------------------------------|---------------------------------|
| Finished water turbidity exceeds target                                         | Supervisor (+ SUMAS if process-related) | As soon as identified           |
| Chlorine residual below minimum                                                 | Supervisor (+ SUMAS if process-related) | As soon as identified           |
| Confirmed positive microbiological result                                       | Supervisor                              | Immediately (same business day) |
| Boil Water Advisory issued ( <i>N/A — no distribution system on this site</i> ) | Follow regulatory authority direction   | As directed                     |
| Chemical spill above what you can safely contain                                | 911 (HAZMAT) + Supervisor + SUMAS       | Immediately                     |
| Power failure > X hours                                                         | Supervisor                              | Per your protocol               |
| Major equipment failure affecting treatment                                     | Supervisor + SUMAS                      | Immediately                     |

**SUMAS Environmental Services Inc. (technical/equipment support): 1-250-374-4151**

**Regulatory reporting hotline:** Not applicable as a discharge-permit requirement (this site has no discharge permit). Confirm with your supervisor whether any other external reporting obligation exists for this site.

## Regulations & Reporting

**⚠ Generic template — likely does not apply to this site:** This section describes the Canadian MUNICIPAL drinking water regulatory framework — Health Canada guidelines, provincial operator certification classes, and distribution-system/THM/HAA reporting requirements aimed at public drinking water utilities. This site is a non-discharge industrial water treatment system with no permit to discharge to a drain, treating contaminated site water for on-site reuse (dust control) only, so these specific municipal drinking-water requirements likely do not apply as written. The actual applicable regulatory framework for this site (if any — e.g. provincial environmental/workplace safety regulations, WorkSafeBC, or Shell Canada internal EHS standards) has not yet been confirmed and should be checked with a supervisor.

### Section 11 — Regulations & Reporting

*You are operating a public utility that directly affects human health. Regulations exist because people got sick when plants were operated carelessly. Understanding the regulatory framework tells you what the minimum standard is — your goal is always to exceed it.*

#### 11.1 The Regulatory Framework (Canadian Context)

*This section uses Canadian references as the default. Update with your specific jurisdiction and permit details.*

##### Federal Level

**Health Canada** sets the *Guidelines for Canadian Drinking Water Quality* — the national science-based benchmarks. These are not legally enforceable by Health Canada but are adopted by provinces as the basis for their regulations.

##### Provincial/Territorial Level

Each province has its own drinking water act and regulations (e.g., Ontario's *Safe Drinking Water Act, 2002*; Alberta's *Water Act*). These set:

- Legal maximum contaminant levels (MCLs)
- Monitoring and reporting requirements
- Operator certification requirements
- Operating permit requirements

##### Your Operating Permit

The single most important document for your plant. It specifies:

- Your plant's specific treatment requirements
- Sampling and monitoring frequencies
- Reporting requirements and deadlines
- Permit limits (your specific MCLs — may be stricter than provincial guidelines)
- Any site-specific conditions

**[ACTION]** Get a copy of your operating permit. Keep it in the control room. Read it cover to cover.

## 11.2 Key Parameters and Limits

The table below shows Health Canada guideline values. Your permit limits may be stricter. Fill in your site-specific limits once you have your permit.

| Parameter                             | Health Canada Guideline      | Your Permit Limit | Operational Target |
|---------------------------------------|------------------------------|-------------------|--------------------|
| Turbidity (treated)                   | ≤ 1 NTU                      | (TBD)             | (TBD)              |
| E. coli                               | 0 CFU/100 mL                 | 0 CFU/100 mL      | 0 CFU/100 mL       |
| Total coliforms                       | 0 CFU/100 mL (treated)       | (TBD)             | 0 CFU/100 mL       |
| Free chlorine residual (distribution) | Maintain measurable residual | (TBD)             | (TBD)              |
| pH                                    | 6.5–8.5                      | (TBD)             | (TBD)              |
| Fluoride                              | ≤ 1.5 mg/L                   | (TBD)             | (TBD)              |
| Trihalomethanes (THMs)                | ≤ 0.1 mg/L (100 µg/L)        | (TBD)             | Minimize           |
| Haloacetic acids (HAAs)               | ≤ 0.08 mg/L (80 µg/L)        | (TBD)             | Minimize           |
| Lead                                  | ≤ 0.005 mg/L (5 µg/L)        | (TBD)             | Minimize           |
| Nitrate (as N)                        | ≤ 10 mg/L                    | (TBD)             | —                  |

## 11.3 Monitoring Requirements

Your permit will specify:

- **What** to test (parameters)
- **Where** to sample (sample points)
- **How often** to sample (continuous, daily, weekly, monthly, annually)
- **Who must analyze** the sample (operator, accredited lab)
- **Holding times** (how fast samples must reach the lab)

### Microbiological monitoring

Bacteriological samples (E. coli, total coliform) must typically be analyzed by a certified/accredited laboratory. These cannot be done in-house without accreditation. Know your lab's name, pickup schedule, and whether they have emergency testing capabilities.

### THM/HAA monitoring

Usually quarterly or annually. Must be sent to an accredited lab. Sample collection and preservation must follow strict protocols (check with your lab — typically requires a special pre-dosed container and 4°C preservation).

## 11.4 Reporting Requirements

### Routine reporting

Most provinces require:

- Monthly operating reports submitted to the regulatory authority
- Annual operating reports
- Records kept for a minimum of 5–10 years (check your permit)

Your monthly report typically includes:

- Flow totals
- Treatment chemical usage
- All water quality test results
- Any permit exceedances with explanation of corrective action
- Maintenance activities

### Adverse event reporting

Certain events trigger mandatory reporting to the regulatory authority:

- Turbidity exceedance above permit limit
- Chlorine residual below minimum
- Any positive E. coli or total coliform result
- Treatment system failure affecting water quality
- Chemical spills
- Equipment failures with water quality implications

**These are typically required within hours of the event — check your permit for timelines.** Late reporting is itself a compliance violation. When in doubt, report immediately.

## 11.5 Operator Certification

### Why certification matters

Water treatment and distribution operators in Canada must hold a valid licence issued by the provincial authority. Operating without a licence (or letting an unlicensed person operate independently) is illegal and puts the utility at risk.

### Certification levels

Most provinces have 4–5 levels based on plant size and complexity:

- **Class 1 / Level 1:** Small, simple plants. Entry point for new operators.
- **Class 2 / Level 2:** Medium-sized plants.
- **Class 3 / Level 3:** Large, complex plants.
- **Class 4 / Level 4 (some provinces):** Very large or complex systems.

Your plant is classified at a specific level — the operator-in-charge must hold a licence at or above that level.

### Getting certified

Requirements vary by province but typically include:

1. Minimum age (usually 18)
2. Application and fee
3. Written examination
4. Work experience (accumulated as a trainee/apprentice under a licensed operator)
5. Continuing education (to maintain certification — e.g., points every 3 years)

**Talk to your supervisor about the certification pathway at your plant.**

### Keeping your certification current

- Track your continuing education hours.
- Renew your licence before it expires — lapsed licences require re-examination in many jurisdictions.
- Any criminal charges or convictions may affect your licence.

## 11.6 Record Keeping Best Practices

### Why records matter:

- Prove to regulators that you were compliant
- Protect YOU personally if there is ever a contamination event or legal action
- Help diagnose recurring problems over time
- Required for accreditation audits

### What to keep:

- All operator log sheets (all shifts, all days)
- All laboratory test results (original, not copies)
- All calibration records for instruments
- All maintenance records
- All chemical delivery and inventory records
- All incident reports and corrective action records
- All regulatory correspondence

**For how long:** Check your permit, but 5–10 years is standard. Some parameters (lead/copper, certain microbiological results) may require longer retention.

**Where to keep them:** Organized, labelled, accessible. If an inspector shows up unannounced, you should be able to produce any record from the last 5 years within 30 minutes.

*Last updated: 2026-06-04 | Source: Initial framework — add your specific permit requirements*

End of manual. Companion documents: LOGBOOK.pdf (daily shift log) and CHEMICAL\_MIXING\_GUIDE.pdf (lime/PACI/floculant mixing & tote mixer steps). Section 12 (Glossary) is maintained separately in the source manual.